

Regulatory Arbitrage and Cross-Border Bank Acquisitions

G. ANDREW KAROLYI and ALVARO G. TABOADA*

ABSTRACT

We study how differences in bank regulation influence cross-border bank acquisition flows and share price reactions to cross-border deal announcements. Using a sample of 7,297 domestic and 916 majority cross-border deals announced between 1995 and 2012, we find evidence of a form of “regulatory arbitrage” whereby acquisition flows involve acquirers from countries with stronger regulations than their targets. Target and aggregate abnormal returns around deal announcements are positive and larger when acquirers come from more restrictive bank regulatory environments. We interpret this evidence as more consistent with a benign form of regulatory arbitrage than a potentially destructive one.

THE RECENT GLOBAL FINANCIAL crisis, caused in part by systemic failures in bank regulation (Levine (2012)), has sparked, among other things, a strong push for both stricter capital requirements and greater international coordination in regulation. For example, 7 of the 10 recommendations of the 2011 Report of the Cross-Border Bank Resolution Group of the Basel Committee for Banking Supervision (BCBS) propose greater coordination of national measures to deal with the increasingly important cross-border activities of banks.¹ A cost

*Karolyi is with Johnson Graduate School of Management, Cornell University and Taboada is with Haslam College of Business, University of Tennessee. We received helpful comments from the Editor, Kenneth J. Singleton, three anonymous referees, Warren Bailey, Leonce Barger, Jim Booth, Alex Butler, David Chor, Stijn Claessens, Werner DeBondt, Larry Fauver, Matt Harris, Jay Hartzell, Tom Jacobs, Jan Jindra, Ed Kane, Oguzhan Karakas, Dong-Wook Lee, Edith Liu, Qingzhong Ma, Dalia Marciukaityte, Yuki Masujima, Pamela Moulton, Luc Renneboog, Stefano Rossi, Klaus Schaeck, Merih Sevilir, Parul Sharma, Joao Santos Silva, Elvira Sojli, Silvana Tenreyro, Neeltje van Horen, Xinli Wang, Tracie Woidtke, and Chu Zhang. We also thank participants at the 2012 EFA meeting (Copenhagen, Denmark), 2011 Asian FMA meetings (Queenstown, New Zealand), Bocconi University's 2012 CAREFIN conference on The Effect of Tighter Regulatory Requirements on Bank Profitability and Risk-Taking Incentives (Milan, Italy), Baruch College, Boston College, Cambridge, Cornell, Cyprus, the Office of the Comptroller of the Currency, DePaul, EPFL-HEC Lausanne, Erasmus, IMF, Ohio State, North Carolina, Pittsburgh, Tennessee, Texas–Austin, Tilburg, and the World Bank. Jonathan Jones, Dilip Patro and their colleagues at the OCC greatly enriched our understanding of bank regulation. Ronnie Chen provided helpful research assistance. All remaining errors are, of course, our own. This paper previously circulated under the title: “The Role of Regulation in Cross-Border Bank Acquisitions: Is It *Really* a ‘Race to the Bottom’?” The authors do not have any conflicts of interest, as identified in the Disclosure Policy.

¹ The group was approved by the BCBS in December 2007, but its report originated with the G20 communiqué of April 2009 and the follow-up G20 Working Group on Reinforcing International

DOI: 10.1111/jofi.12262

of centralizing bank regulation across countries is that it limits cross-country regulatory competition to the extent that a fully harmonized global regulation imposes uniform standards across countries (Acharya (2003), Dell'Ariccia and Marquez (2006)). A benefit, however, is that it internalizes any interdependencies that exist across countries due to the integration of their financial systems. Indeed, the push for stricter regulations has been driven in large part by concerns about one such interdependency, namely, an increase in the risk of “regulatory arbitrage,”² whereby banks from countries with strict regulations engage in cross-border activities in countries with weaker regulations.

There are two views on the consequences of regulatory arbitrage. On the one hand, banks engaging in regulatory arbitrage can maximize shareholder value and improve capital allocation if such activities occur when banks are constrained from pursuing profitable investment opportunities due to costly regulations in their home country. These benefits may accrue not just to the acquirer, but also to the target, which may benefit from “bonding” to a more robust regulatory regime after being acquired by banks from countries with stronger supervision.³ On the other hand, banks engaging in regulatory arbitrage can pursue value-destroying activities through excessive risk-taking, for example, by acquiring targets in countries with lax regulations and weak supervision. This form of regulatory arbitrage could have adverse consequences for bank performance and shareholder value, not only for the parties to the deal but also for the banking system as a whole, and could even be a catalyst for a harmful “race to the bottom” in bank regulations.⁴ Regulatory arbitrage of this harmful form may be especially dangerous as it can increase the fragility of interconnected financial systems around the world if the acquiring banks can extract subsidies from the host country’s regulator, central bank, or taxpayers for losses from its more weakly monitored risk exposures.⁵

Cooperation and Promoting Integrity in Financial Markets, which became a permanent initiative in the form of the Financial Stability Board. The BCBS, as a forum for regular cooperation on banking supervisory matters, has been in existence since 1974 and reached prominence with the Basel Capital Accords in 1988.

² BCBS member José María Roldán stated at the Asian Banker Summit in Hong Kong in April 2011, “If we have higher capital requirements, we are going to have higher incentives for regulatory arbitrage. Within banks, across banks, across countries, if you have an uneven application of Basel III you will see banking activity going to the country that has a softer approach.”

³ The bonding hypothesis (Coffee (1999), Stulz (1999)) has been widely documented in the cross-listing literature (see, e.g., Doidge, Karolyi, and Stulz (2004)). Bonding could happen through tougher discipline imposed by stronger regulatory authorities from the acquirer’s home country for the newly consolidated entity. This is similar to bonding in the cross-border M&A literature whereby the target firm usually adopts the governance structures of the country of the acquiring firm (Rossi and Volpin (2004), Bris and Cabolis (2008), Ellis et al. (2012), Starks and Wei (2013)).

⁴ By engaging in cross-border bank acquisitions in countries with weaker regulations, acquirers may also extract safety-net subsidies from home country regulators. We thank Edward Kane for providing this alternative interpretation.

⁵ Acharya (2003) models how international convergence of bank capital requirements can lead to an unintended race-to-the-bottom equilibrium if not accompanied by consistent resolution policies across regimes. Banks that operate across borders will undertake greater risk in more forbearing regimes, which reduces bank profits in less forbearing regimes and forces those banks to exit. All

In this paper, we shed light on the motives behind regulatory arbitrage by examining one of the most important types of investment decisions that banks can make—cross-border acquisitions. Cross-border deals are a particularly useful setting to evaluate the effects of regulatory restrictions *not only* because the acquiring banks can escape some of the regulatory restrictions in their home country by acquiring institutions in weaker regimes *but also* because of the enormous growth in bank consolidation—domestic and especially cross-border—facilitated by major regulatory changes around the world.⁶ We increase the power of our tests by controlling for other motives for bank acquisitions, such as efficiency, market power, as well as governance-related motives, and by benchmarking against purely domestic deals. To the best of our knowledge, ours is the first study to examine regulatory arbitrage in cross-border bank acquisitions on a global basis.

Our sample of 7,297 domestic and 916 majority cross-border deals is cumulatively valued in excess of \$2.8 trillion involving acquirers and targets from 78 countries over the period 1995 through 2012. We first evaluate how differences in regulations influence the volume of cross-border bank acquisitions and the flow of deal activity between the home countries of the bank acquirers and targets to determine whether the flows are in line with regulatory arbitrage. We next examine the impact of differences in bank regulation on shareholder wealth as measured by the short-run stock price reactions of the acquirer and target banks to deal announcements. Finally, we examine how changes in merger control policy influence both cross-border bank acquisitions and the potential impact of differences in bank regulations on those flows.

Cross-border acquisitions are, of course, just one way in which banks can engage in regulatory arbitrage. But examining such deals allows us to disentangle whether regulatory arbitrage activities represent the pursuit of excessive risk-taking or the more benign escape from costly regulations by controlling for deal- and bank-specific attributes in addition to country-level factors. We hypothesize that, for banks that are more likely to engage in value-destroying regulatory arbitrage through cross-border acquisitions (the race-to-the-bottom view), we should observe an adverse market reaction to cross-border acquisitions from good to weak countries, all else being equal.⁷ In addition, to the extent that regulatory arbitrage of this form drives cross-border bank

regulators therefore converge on the worst level of forbearance, which destabilizes the banking system. Similar models include Dell' Ariccia and Marquez (2006), Morrison and White (2009), and Agarwal et al. (2012). Acharya, Wachtel, and Walter (2009) discuss how regulatory arbitrage activities might expose all jurisdictions to the influence of excess risk-taking.

⁶ Examples include the 1999 Gramm-Leach-Bliley Financial Services Modernization Act in the United States that overturned the Glass Steagall Act of 1933, which had separated commercial from investment banking activities; the Federal Reserve's Regulation K that reduced the regulatory burden on foreign banks operating in the United States (final amendment in October 2001); and the 1989 Second Banking Directive in the European Union (EU) that created a single banking license valid throughout the EU.

⁷ Although a race to the bottom is the outcome of regulatory choices associated with the harmful motive for cross-border bank acquisitions, for simplicity we use the term "race to the bottom" to refer to the harmful form of regulatory arbitrage.

acquisitions, we should observe both “push” and “pull” factors at work. That is, acquirer banks should be influenced as much by tough regulatory restrictions of their home regime as by the weak regulatory restrictions of the target regime. In contrast, for banks engaging in the benign form of regulatory arbitrage through cross-border acquisitions, we should observe a positive market reaction for the target and possibly also acquirer bank due to the former bonding to a tougher regulatory regime. In this case, push factors should influence cross-border flows more intensely.

We uncover several new results. First, cross-border bank acquisitions are indeed more likely to involve acquirers from countries with stronger supervision, more restrictions on bank activities, and more stringent capital requirements, which suggests that regulatory arbitrage is a motive for cross-border bank acquisitions. These factors are important even after controlling for broader measures of corporate governance and a host of other factors previously shown to influence cross-border acquisition flows. Second, target banks’ (and aggregate) abnormal returns are positive *and* significantly larger when acquirers are from countries with more restrictions on bank activities, stricter capital requirements, stronger private monitoring, and better overall regulatory quality. This finding adds support to the benign escape-from-costly-regulations or bonding views of regulatory arbitrage. Finally, we document that our results on cross-border bank acquisition flows are driven primarily by push factors. Acquirers from tough regimes do not necessarily target countries with the weakest regulatory quality, a finding that provides additional support to the benign view of regulatory arbitrage.

We address concerns about potential endogeneity and reverse causality in our tests by employing instrumental variables for differences in bank regulations and their changes over time and by accounting for unobserved time-invariant country characteristics that may influence cross-border acquisition flows. None of these solutions can completely eliminate concerns that regulations may endogenously respond to changes in global capital market flows, but our key findings are resilient to these alternative identification approaches. Our key results are also robust to different bank samples, subperiods, measures of bank regulation, estimation methods, and combinations of firm-, deal- and country-level controls.

We contribute to several strands of the finance literature. Cross-border studies of bank regulation show that tough regulatory restrictions on bank activities and barriers to foreign entry hurt banking sector performance (Barth, Caprio, and Levine (2006)). In a more recent study, Laeven and Levine (2009) find that tougher bank regulation reduces banks’ risk-taking behavior, although the impact of regulations on risk-taking depends critically on each bank’s ownership structure. Our study on cross-border bank acquisitions contributes to this stream of research by providing unique evidence on the potential economic consequences of *changes* in bank regulation. Cross-border acquisitions are one mechanism through which banks can change their regulatory environment and potentially engage in regulatory arbitrage by acquiring a majority stake in a

target from a much weaker regime.⁸ As a result, this cross-border setting allows for an experiment with rich variation in the sign *and* magnitude of changes in regulatory constraints experienced across the cross-border deals we study.

Our study also contributes to the more recent literature examining regulatory arbitrage and the need for global coordination in financial regulation. Houston, Lin, and Ma (2012) examine international bank flows and find evidence of regulatory arbitrage, as banks tend to predominantly transfer funds to countries with fewer regulations. They argue that the direction of flows could signal a harmful race to the bottom. However, they also find that flows tend to go to countries with stronger property and creditor rights. In a related paper, Ongena, Popov, and Udell (2013) examine the impact of home country regulations on lending activity abroad by European banks with a presence in 16 Eastern European countries. They find that banks from countries with tighter restrictions on bank activities and more capital requirements tend to make riskier loans abroad, in line with the race-to-the-bottom view of regulatory arbitrage, although stronger supervision at home reduces risk-taking abroad.

Examining cross-border acquisitions, we contribute to this literature by providing more direct evidence of the type of regulatory arbitrage that may be taking place. We are able to do so *at the deal level* by evaluating the short- and long-run consequences of the deals in terms of the share price reactions of the target and acquirer banks. Furthermore, while the evidence in Houston, Lin, and Ma (2012) is based on bank flows from 26 mostly OECD source countries and this primarily captures bank flows that originate in developed countries,⁹ our analysis incorporates a much larger number of acquiring countries and hence captures deals involving acquirers from many emerging markets with different levels of regulatory quality. The role of banks from emerging markets as acquirers (and the extent to which they engage in regulatory arbitrage) is important, especially because of the increase in cross-border bank acquisition activity involving acquirers from emerging markets that occurred during the latter half of our sample period. By studying cross-border bank acquisitions, we can also more directly observe whether banks are engaging in such deals to expand their current activities, or to diversify into other (perhaps riskier) activities that may be restricted in their home countries.

Two recent studies do evaluate regulatory issues in cross-border banking mergers, but focus on the context of the European Union (EU). Carbo-Valverde,

⁸ While the home country supervisor retains ultimate supervisory authority over the consolidated operations of the acquiring bank, monitoring bank activities in host countries is complicated by several factors such as distance and lack of coordination with host country supervisors. In addition, by acquiring banks in countries with fewer restrictions on bank activities, acquirers may engage in activities prohibited in the home country (e.g., providing insurance services), further complicating the home country supervisor's role.

⁹ Ongena, Popov, and Udell (2013) have a limited number of parent banks in their study, predominantly from Western Europe. Unlike our study, they examine the direct lending activities of these banks abroad. However, our study can capture potential regulatory arbitrage involving banks from emerging markets. We can also assess other avenues of regulatory arbitrage—such as insurance and investment banking services—that banks may engage in by entering into diversifying cross-border acquisitions.

Kane, and Rodriguez-Fernandez (2012) find evidence of risk-shifting behavior around 165 EU deals between 1993 and 2004, which they interpret as consistent with the race-to-the-bottom form of regulatory arbitrage. In contrast, Hagendorff et al. (2012) find evidence inconsistent with the race-to-the-bottom form for a sample of domestic and cross-border EU deals between 1997 and 2007. Neither study examines how cross-border bank acquisition flows relate to differences in regulations.¹⁰ Finally, a contemporaneous study by Dong, Song, and Tao (2011) examines more than 2,000 cross-border bank mergers completed between 1990 and 2007 and finds evidence of the harmful form of regulatory arbitrage. Unlike our study, which focuses on majority cross-border bank acquisitions, their analysis focuses on the aggregate number of cross-border deals and does not take into account time-varying differences in regulatory quality.

We also contribute to the literature on cross-border bank acquisitions by exploring differences in bank regulation as a motive for the increase in cross-border bank acquisition activity over the past few years. Despite the many potential gains from cross-border bank acquisitions (Cornett and Tehranian (1992), Berger, Hunter, and Timme (1993), Pilloff and Santomero (1998)), there is little empirical support for the argument that banks engaging in such deals realize cost or profit efficiencies (Vander Vennet (2003), Amel et al. (2004), Correa (2009)). Many studies argue that there exist barriers that prevent the proliferation of cross-border bank deals and impede the full exploitation of potential synergies in such mergers (Berger, De Young, and Udell (2001), Focarelli and Pozzolo (2001), Buch and DeLong (2004)).¹¹ Despite the failure to uncover gains in cross-border bank acquisitions, only a couple of studies examine the role that bank regulation or corporate governance might play in such deals.¹²

The rest of the paper is organized as follows. In Section I, we describe the data. In Section II, we examine the determinants of cross-border bank acquisition activity. In Section III, we present results on the stock price reaction to cross-border bank acquisition announcements. In Section IV, we examine the impact of merger control policies on cross-border bank acquisition activity. In Section V, we present results from robustness tests. We conclude in Section VI.

I. Data and Summary Statistics

To explore the determinants of cross-border bank acquisitions, we build a broad sample of domestic and cross-border bank acquisitions using data from

¹⁰ Buch and DeLong (2008) find that regulatory factors play a minor role in explaining the number of bank mergers between countries. Banks from countries with fairly priced deposit insurance tend to reduce risk after a merger.

¹¹ Campa and Hernando (2006) find positive excess returns for targets in cross-border deals, although the returns tend to be lower than for targets in domestic mergers. Amihud, DeLong, and Saunders (2002) find significantly negative abnormal returns for the acquirers. Cybo-Ottone and Murgia (2000) find that cross-border deals do not capture positive expectations from the market.

¹² An exception is Hagendorff, Collins, and Keasey (2008), who analyze the share price effects of 31 cross-border bank acquisitions in Europe and the United States. They document an inverse relationship between the quality of legal protections for minority investors in the target bank's country and the bidder's abnormal returns.

Thomson's Securities Data Corporation (SDC) Platinum database. A bank acquisition is defined as one in which the acquirer is a commercial bank or bank holding company and in which targets may also be insurance companies, mortgage bankers, and security brokers. In line with the literature, we exclude privatizations, leveraged buyouts, spin-offs, recapitalizations, exchange offers, repurchases, and self-tender offers. We also drop 23 deals in which SDC reports the target and the acquirer to be the same entity, although the deal is not classified as a share repurchase. The initial sample consists of 15,942 bank acquisitions announced between January 1, 1995 and December 31, 2012. Because we are interested in deals involving changes in control, we focus on majority acquisitions in which the acquirer owns less than 50% of the target's stock before the deal and more than 50% of the target's stock after the deal. This results in a sample of 8,554 majority deals totaling \$2.84 trillion, including 1,132 cross-border deals (totaling \$360 billion). We next exclude deals involving countries with no available information on banking regulation (to be discussed below) and those with missing data on our main country-level variables.¹³ Our final sample consists of 7,297 (916) domestic (cross-border) majority acquisitions covering 78 countries with a total value of \$2.5 trillion (\$342 billion).

Table I provides descriptive statistics for the sample. To get stock price information, we merge this sample with the Thomson Reuters DataStream and CRSP databases. Panel A shows considerable variation across years in the number of domestic and cross-border acquisitions. The number of completed cross-border deals is lowest (highest) in 1995 (2006) at 28 (94) deals, while the total value of completed deals is lowest (highest) in 2003 (2000) at \$5.1 billion (\$62.4 billion). The number (value) of cross-border deals as a fraction of all completed deals rises from 3.9% (3.2%) in 1995 to 11.7% (19.9%) in 2012. Cross-border deals are typically larger than domestic deals (\$773 million compared to \$693 million).

Panel B of Table I shows that, after merging the initial SDC sample with the DataStream/CRSP databases (our sources for bank stock price information), the sample size drops to 3,589 (621) completed domestic (cross-border) deals, out of which 2,481 (336) domestic (cross-border) deals report value. For this sample, we collect accounting information on targets and acquirers from Thomson Reuters's WorldScope and additional data from Bloomberg. As expected, the average deal value in the merged SDC and DataStream/CRSP sample is significantly larger than in the original SDC sample. The average value of domestic (cross-border) deals is \$889 (\$933) million in the merged sample, compared to \$693 (\$773) million in the original SDC sample.

¹³ We lose 216 majority cross-border deals (19.1% of the original sample) with a cumulative value of \$18.2 billion from countries with missing data on banking regulation or other country-level variables. Deals involving institutions (as targets or acquirers) from the following 15 countries are excluded because of lack of data on banking regulation: American Samoa, Andorra, Bahamas, Barbados, Bermuda, Brunei Darussalam, Faroe Islands, Greenland, Mauritania, Monaco, Netherlands Antilles, Reunion, San Marino, Uzbekistan, and Zaire. We also drop the following countries without any majority cross-border acquisition activity during our sample period: Belize, Côte d'Ivoire, Jamaica, Kyrgyz Republic, Myanmar, Tajikistan, Zambia, and Zimbabwe.

Table I
Bank Acquisitions around the World

This table describes all majority acquisitions in which the acquirer is a commercial bank or bank holding company, while targets may also be insurance companies, mortgage bankers, and security brokers. The year represents the year in which the deal was announced. Acquisitions in which the target institution or the acquiring institution's country of origin was not identified are excluded. The initial sample, broken down by year in Panel A, consists of all bank majority acquisitions (those in which the acquirer owns more than 50% of the target after the deal) announced between January 1995 and December 2012. Data are obtained from Thomson's SDC Platinum database. All recapitalizations, spinoffs, leveraged buyouts, divestitures, share repurchases, and privatizations are excluded. We then construct a sample of domestic and cross-border bank acquisitions with stock price information available in the Thomson Reuters DataStream and CRSP databases (SDC + DataStream/CRSP sample in Panel B). The SDC + DataStream/CRSP sample consists of all deals for which we can compute abnormal returns for either acquirers or targets. # denotes the number of deals. *P*-values (in parentheses) are for *t*-tests (Kolmogorov-Smirnov nonparametric tests) for differences in mean (median).

Panel A: Full Sample by Year											
Year	Domestic Acquisitions						Cross-Border Acquisitions				
	Total Announced	# Completed	Total Value (\$Million)	Mean Value (\$Million)	# Reporting	Value	Total Announced	# Completed	Total Value (\$Million)	Mean Value (\$Million)	Reporting Value
1995	691	689	\$193,426	\$632	306	28	28	28	\$6,459	\$461	14
1996	521	519	\$53,955	\$203	266	31	31	31	\$7,392	\$493	15
1997	516	515	\$196,590	\$612	321	60	60	59	\$15,306	\$567	27
1998	546	543	\$335,128	\$989	339	48	48	48	\$20,806	\$771	27
1999	484	480	\$296,030	\$1,130	262	48	48	47	\$18,597	\$1,162	16
2000	426	424	\$216,026	\$969	223	72	72	70	\$62,353	\$1,834	34
2001	385	385	\$86,740	\$432	201	61	61	60	\$15,357	\$495	31
2002	373	372	\$46,044	\$276	167	41	41	41	\$23,503	\$1,383	17
2003	380	375	\$125,190	\$577	217	45	45	43	\$5,111	\$183	28
2004	383	379	\$166,079	\$765	217	51	50	50	\$33,367	\$1,335	25
2005	363	361	\$142,517	\$713	200	71	71	69	\$6,009	\$223	27
2006	347	344	\$185,813	\$1,038	179	96	96	94	\$36,157	\$709	51
2007	333	329	\$114,541	\$674	170	72	72	70	\$39,095	\$889	44
2008	261	261	\$176,440	\$1,634	108	45	45	45	\$21,776	\$1,037	21
2009	322	318	\$38,670	\$439	88	37	37	37	\$6,224	\$445	14

(Continued)

Table I—Continued

Panel A: Full Sample by Year									
Domestic Acquisitions					Cross-Border Acquisitions				
Year	Total Announced	# Completed	Total Value (\$Million)	Mean Value (\$Million)	# Reporting Value	Total Announced	# Completed	Total Value (\$Million)	Mean Value (\$Million)
2010	384	384	\$40,567	\$441	92	33	32	\$12,212	\$678
2011	347	345	\$26,913	\$228	118	47	46	\$6,348	\$334
2012	235	234	\$22,889	\$286	80	31	31	\$5,700	\$407
TOTAL	7,297	7,257	2,463,561	\$693	3,554	916	901	\$341,772	\$773

Panel B: Sample Comparison: Completed Majority Acquisitions									
Initial SDC Sample					SDC + DataStream/CRSP Sample				
	Domestic	Cross-Border	Difference (p-value)		Domestic	Cross-Border	Difference (p-value)		
Mean value (\$Million)	\$693	\$773	(0.500)		\$889	\$933	(0.776)		
Total value (\$Million)	\$2,463,561	\$341,772			\$2,206,688	\$313,598			
Total deals announced	7,297	916			3,607	629			
Total deals completed	7,257	901			3,589	621			
# Reporting value	3,554	442			2,481	336			
Test of difference with SDC sample (p-value)					(0.000)	(0.000)			

Our sample of bank acquisitions is geographically diverse comprising a total of 78 countries. Table II reports the number and value of cross-border and domestic majority bank acquisitions for the top 25 target and acquiring countries based on the value of cross-border bank acquisition activity. The results show that the United States was the most active target market in cross-border acquisitions during our sample period (148 deals totaling \$145 billion), followed by the United Kingdom (73 deals totaling \$51 billion) and Germany (41 deals totaling \$15 billion). Many target banks are from several emerging markets including Brazil, Ukraine, and Argentina. On the acquisition side, banks from Spain were the most active acquirers in cross-border deals during our sample period (54 deals totaling \$54 billion), followed by the United Kingdom and Germany, which both have higher counts but lower total values than Spain. Banks from China and Brazil were also active acquirers during our sample period. In the Internet Appendix, we evaluate the full matrix of bank acquisitions by acquirer-target country pair for the top 25 target and acquirer countries, respectively.¹⁴ We learn that Spain, the most aggressive acquirer country (\$54 billion) over the sample period, concentrated its attention on the United States (\$21 billion) and the United Kingdom (\$19 billion), but also acquired target institutions in several Latin American countries, such as Brazil and Chile. U.K. banks were active acquirers in more diverse countries, such as the United States (\$36 billion), Hong Kong, Australia, Ireland, and Russia. The United States, the largest target market, attracted most interest from the United Kingdom and Spain, but also attracted interest from Canada (\$28 billion), Germany (\$19 billion), the Netherlands, and Switzerland. Interestingly, most of these countries have lower regulatory quality than the United States, based on many of our regulatory measures (see Appendix A). Target banks from several emerging countries, including Brazil, Ukraine, and Argentina, are well represented in our sample.

To examine whether the quality of banking system regulation of the target or acquirer country has any influence on cross-border acquisition flows, target choices, and share price reactions to acquisition announcements, we first use four measures of the quality of bank regulation from Barth, Caprio, and Levine (2013). These measures, which are built from four surveys sponsored by the World Bank and conducted in 1999, 2003, 2007, and 2011, include:¹⁵ *Restrictions on bank activities*, an index that measures regulatory impediments to

¹⁴ Note that the top 25 acquirer and target countries taken as a group of countries account for 93.9% of the total value of cross-border deals in our sample. These tables are provided in the Internet Appendix, available in the online version of the article on the Journal of Finance website.

¹⁵ The surveys comprise 12 parts with about 175 questions addressed to regulators in various countries covering ownership, capital, bank activities, external audit requirements, internal management requirements, liquidity/diversification requirements, depositor protection schemes, accounting/disclosure requirements, discipline, and supervision. In results reported in the Internet Appendix, we explore many of the indexes available, but, in our main analysis, we focus on the four indexes that capture not only regulations stressed by the BCBS, but also those that theory and empirical evidence highlight as affecting bank risk-taking behavior and influencing the stability of the banking system.

Table II
Completed Majority Domestic and Cross-Border Bank Acquisitions
by Target and Acquirer Countries, 1995–2012

This table reports descriptive statistics on all completed domestic and cross-border majority bank acquisitions with available information on the total value of the deal. Bank acquisitions are defined as those in which the acquirer is a commercial bank or bank holding company, while targets may also be insurance companies, mortgage bankers, and security brokers. The deals are listed by country of origin of the target and acquirer. The data are obtained from Thomson's SDC Platinum database for all acquisitions announced and completed between 1995 and 2012. Reported values are in constant (2012) U.S. dollars and the number of deals is indicated by #.

	Target					Acquirer							
	Cross-Border Deals		Domestic Deals		All Deals	Cross-Border Deals		Domestic Deals		All Deals			
	#	Value (\$Million)	#	Value (\$Million)	Total Value (\$Million)	#	Value (\$Million)	#	Value (\$Million)	Total Value (\$Million)			
United States	148	\$145,215	4,740	\$1,409,081	4,888	\$1,554,296	Spain	54	\$54,443	134	\$60,866	188	\$115,309
United Kingdom	73	\$50,997	97	\$101,012	170	\$152,009	United Kingdom	75	\$47,723	97	\$101,012	172	\$148,735
Germany	41	\$14,519	192	\$40,619	233	\$55,138	Germany	78	\$39,535	192	\$40,619	270	\$80,155
Turkey	21	\$13,199	20	\$989	41	\$14,188	Canada	39	\$31,845	65	\$7,176	104	\$39,021
Finland	11	\$12,318	34	\$4,033	45	\$16,351	Switzerland	55	\$26,181	94	\$42,904	149	\$69,085
Brazil	22	\$11,719	74	\$28,873	96	\$40,593	United States	82	\$26,003	4,740	\$1,409,081	4,822	\$1,435,085
Austria	10	\$10,653	20	\$993	30	\$11,646	France	63	\$19,353	97	\$32,395	160	\$51,748
Italy	28	\$8,280	286	\$201,630	314	\$209,910	Netherlands	34	\$11,874	29	\$1,424	63	\$13,298
Ukraine	27	\$7,671	12	\$0	39	\$7,671	Belgium	38	\$11,490	21	\$22,508	59	\$33,997
Hong Kong	20	\$7,364	17	\$2,541	37	\$9,904	Denmark	14	\$8,757	30	\$4,242	44	\$12,999
Russia	29	\$5,870	193	\$2,342	222	\$8,212	Sweden	28	\$8,398	18	\$3,246	46	\$11,643
Portugal	14	\$5,835	16	\$3,511	30	\$9,346	Austria	26	\$7,080	20	\$993	46	\$8,074

(Continued)

banks engaging in securities market activities (underwriting, brokering, dealing, mutual funds), insurance activities (underwriting and selling), and real estate (development or management); *Stringency of capital regulation*, an index measuring how much capital banks must hold, as well as the sources of funds that count as regulatory capital; *Official supervisory power*, an index that measures whether supervisory authorities have the power to take actions to prevent or correct problems; and *Private monitoring*, an index that measures whether there are incentives for the private monitoring of firms.¹⁶ In addition, we use a composite index of the strength of bank regulation, *Regulation overall*, which is the first principal component of the four indices from Barth, Caprio, and Levine (2013).¹⁷ Because the indices are not available annually, we use the value of the variables from the first survey (data as of 1999) for the period 1995 to 2001, the value of the variables from the second survey (data as of 2002) for the period 2002 to 2004, the value of the variables from the third survey (data as of 2005) for the period 2005 to 2010, and the value of the variables from the last survey for the period 2011 to 2012. These and other variables used in our analyses are described in detail in Appendix B. Appendix A presents the regulatory index values by country.

We also consider several measures of country-level governance and development that have been shown to influence cross-border acquisition activity. Our primary measure of governance is *Governance index*, which is the average of the six governance indicators of Kaufmann, Kraay, and Mastruzzi (2009): voice and accountability, regulatory quality, political stability, government effectiveness, rule of law, and control of corruption. To control for financial development and growth, we use the log of Gross Domestic Product (GDP) per capita (*Log GDP per capita*) and growth in real GDP (*GDP growth*) obtained from the World Bank's World Development Indicators database. We also control for *Real stock market return* and *Real exchange rate return*, which have been shown to be important determinants of cross-border mergers and acquisitions (Erel, Liao, and Weisbach (2012)). To control for the size of the banking sector, we include total credit provided by deposit money banks to the private nonfinancial sector (*Bank credit*), and to control for the composition of the banking sector, we include assets of the top three banks as a proportion of all commercial bank assets (*Bank concentration*). These two measures are obtained from the Bank for International Settlements (BIS) and the World Bank's Global Financial

¹⁶ As an example, consider the *Stringency of capital regulation* index, which is based on answers to a series of nine questions that include: Is the minimum capital asset ratio requirement risk weighted in line with the Basel guidelines, and does the minimum ratio vary as a function of market risk? The index measures the regulatory approach to assessing and verifying the degree of capital at risk in a bank. The *private monitoring* index determines whether a certified audit is required, the percent of the 10 biggest banks that are rated by international agencies, whether income statements include accrued/unpaid interest or principal on nonperforming loans, and whether a deposit insurance scheme exists.

¹⁷ We extract the first principal component from the index values from each of the four surveys. The proportion of the variance accounted for by the first principal component ranges from 32% to 38%, depending on the survey.

Development Database (Beck, Demirgüç-Kunt, and Levine (2009), Čihák et al. (2012)), respectively.

Several variables are commonly used in the trade literature to explain resistance to greater cross-border trade flows, which we obtain from Mayer and Zignago (2011).¹⁸ These include the log of the circle distance in kilometers between countries' capitals (*Distance*) and indicator variables for countries that share the same language (*Same language*), that share a border (*Contiguous*), and that have had a colonial link (*Colony*). We also obtain bilateral trade data from the International Monetary Fund's (IMF) Direction of Trade Statistics to measure the maximum of bilateral exports and imports between countries i and j as a fraction of the total imports by country j (*Bilateral trade*) and thereby determine whether differences in regulation can explain cross-border acquisition flows over and above those influenced by trade. Finally, we incorporate a measure of exogenous growth opportunities (*Growth opportunities*, the log of the inner product of the vector of global industry price-to-earnings (PE) ratios and the vector of country-specific industry weights) from Bekaert et al. (2007) to capture additional factors that may affect cross-border deals and help mitigate endogeneity concerns.¹⁹ Appendix C provides descriptive statistics and a correlation matrix for all the country-level variables.

II. Determinants of Cross-Border Bank Acquisition Activity

A. Cross-Sectional Determinants of Cross-Border Bank Acquisitions

We first examine how differences in regulation influence cross-border bank acquisitions using a multivariate regression framework. We arrange our data set to produce a matrix of (69×68) ordered country pairs.²⁰ For each country pair, we compute *Cross-border ratio* as the total number of majority cross-border bank acquisitions in which the target is from country i and the acquirer is from country j ($i \neq j$) as a proportion of all bank acquisitions (domestic and cross-border) in target country i during the sample period. Consistent with other studies of cross-border mergers and acquisitions, we include domestic acquisitions in the denominator of *Cross-border ratio* to implicitly control for factors that may influence both domestic and cross-border acquisitions.

We apply a gravity model to explain *Cross-border ratio* as a function of various country characteristics, measured as differences between acquirer and target countries. First, we control for differences in bank regulation using the five

¹⁸ We are motivated by studies of the gravity equation model for trade in which exports from country i to country j are proportional to the product of the two countries' GDPs and are inversely proportional to factors that might create trade resistance. See, among others, Eaton and Kortum (2002) and Anderson and Wincoop (2003). Other banking studies that apply the gravity equation include Blank and Buch (2007) and Herrmann and Mihaljek (2013).

¹⁹ Because this measure of growth opportunities does not use local price information, Bekaert et al. (2007) argue that it can be useful in addressing endogeneity problems.

²⁰ Our sample drops to 69 countries after eliminating countries without targets involved in cross-border acquisitions during the sample period.

indices of bank regulation: (i) *Restrictions on bank activities*, (ii) *Stringency of capital regulation*, (iii) *Official supervisory power*, (iv) *Private monitoring*, and (v) *Regulation overall*. Next, we control for factors previously shown to influence cross-border acquisitions. These controls include *Governance index*, *Log of GDP per capita*, *GDP growth*, *Bank concentration*, *Bank credit*, *Real stock market return*, *Real exchange rate return*, *Distance*, *Bilateral trade*, *Growth opportunities*, *Same language*, *Contiguous*, and *Colony*. In these cross-sectional tests, we average all measures across the full sample period. In the next part of the analysis, we extend the cross-sectional analysis across country pairs to a panel setting across country-pair-years.

It is possible that an increase in cross-border activity induces the very changes in banking regulations that we seek to study, a form of reverse causality that can impact interpretation of the findings. We attempt to address such concerns by instrumenting our main regulatory variables of interest. In particular, we use several variables that empirical and theoretical literature suggest could shape banking regulations and institutions but that are not expected to have a first-order effect on cross-border bank acquisition activity. First, we use the government ownership of banks in 1970 from La Porta, Lopez-de-Silanes, and Shleifer (2002) as an instrument. A government's direct involvement in the banking sector of a country may help shape bank regulations, but the level of government ownership of banks in 1970 should not have a direct effect on the flows in cross-border bank acquisitions during our sample period. Second, we use the number of systemic banking crises in a country during the 1970s and 1980s (from Laeven and Valencia (2012)) as an instrument. As evidenced by the recent global financial crisis, banking crises tend to be followed by a push for stricter regulations. Countries with more systemic banking crises may have undergone significant changes in their regulatory environment. While the number of banking crises in a country may discourage the flow of cross-border bank acquisitions, crises as far back as the 1970s should not have a first-order effect on cross-border bank flows during our sample period. Finally, following Beck, Demirgüç-Kunt, and Levine (2006), we use the percentage of years since 1776 that a country has been independent as an instrument. Countries that have been independent for a longer period of time may have been able to adopt regulations that are more beneficial for economic development.

Because our main variables of interest are differences in the quality of bank regulation between acquirer and target countries, in the first stage, we regress the differences in the regulatory quality variables on the differences in the three instruments between the acquirer and target countries, including the full vector of controls used in the second-stage regressions. Results from these first-stage regressions are available in the Internet Appendix. While no instrument is perfect, these three instruments (measured as differences between the acquirer and target countries) appear to exhibit significant explanatory power for differences in bank regulation in the 2000s without having a first-order

effect on our dependent variable.²¹ We use Hansen's J -statistic overidentification test (χ^2) of the joint null hypothesis that the instruments are valid. We reliably reject the null in all of our specifications.

B. Poisson Pseudo-Maximum Likelihood and Simulated Method of Moments

Our dependent variable in these cross-sectional regressions has a large proportion of zeros as there is no cross-border acquisition activity between many country pairs.²² As a result, a two-stage least squares (2SLS) approach may produce biased estimates. We therefore use two alternative approaches commonly employed in the trade literature to address the prevalence of zero values.²³ Our primary approach uses the Poisson pseudo-maximum likelihood (PPML) estimator proposed by Santos Silva and Tenreyro (2006) and tested in several studies that estimate gravity equations (Tenreyro (2007), Fally (2013), Irarrazabal, Moxnes, and Oromolla (2013)).²⁴ In our case, the cross-border acquisition flows are projected on the exponentiated product of the coefficients and independent variables, and the coefficients are estimated using generalized method of moments (GMM). An instrumental variables approach is easily accommodated in PPML and can satisfy conditions to produce a consistent estimator (Tenreyro (2007)). The PPML estimator has been shown to perform well compared to other approaches in the presence of a large proportion of zeros (Santos Silva and Tenreyro (2011)).

To explain the derivation of the PPML estimator, we follow Tenreyro (2007) and state our cross-border flows equation in its exponential form to simplify notation

$$CB_{i,j} = \exp(X_{i,j}\beta) + \varepsilon_{i,j}, \quad (1)$$

where $CB_{i,j}$ is the *cross-border ratio*, or the total number of majority cross-border deals in which the target is from country i and the acquirer is from country j ($i \neq j$), scaled by the total number of domestic and cross-border bank acquisitions in target country i , and $X_{i,j}$ is the vector of controls described

²¹ In additional robustness tests reported in the Internet Appendix, we use instruments based on Hofstede's (1980) cultural dimensions: power distance, masculinity, uncertainty avoidance, and individualism. These are well-established measures of a country's culture that should not have a first-order effect on flows of acquisition activity. The results using these alternate instruments are in line with those reported in our main tables. We also conduct robustness results with instruments suggested by Houston, Lin, and Ma (2012) including ethnic fractionalization, the average Gini coefficient as a measure of income inequality, and a common law indicator for the legal origin of a country's commercial laws.

²² Of the 4,692 observations in our matrix of (69×68) ordered country pairs, 91.97% have zero values.

²³ In the trade literature, the problem is compounded by the fact that the gravity equation is estimated using logs, which causes problems in inference when the data contains many zeros, given that these must be dropped. We do not use logs and therefore we do not drop zeros from our regressions.

²⁴ PPML does not require the data to follow a Poisson distribution, which is why it is described as a pseudo-maximum likelihood estimator and not a maximum likelihood estimator.

earlier, measured as differences between the acquirer and target countries. The PPML estimator is given by

$$\hat{\beta} = \arg \max \sum_{i,j}^n [CB_{i,j} \times (X_{i,j}b) - \exp(X_{i,j}b)]. \quad (2)$$

Thus, the PPML estimator solves for the following first order condition:

$$\sum_{i,j}^n [CB_{i,j} - \exp(X_{i,j}\hat{\beta})] CB_{i,j} = 0. \quad (3)$$

As explained by Santos Silva and Tenreiro (2006), all that is required for this estimator to be consistent is for the conditional mean to be correctly specified. Thus, the PPML estimator is a GMM estimator that solves the moment conditions in the above equation. Unlike the nonlinear least squares estimator, which gives more weight to large observations, which are typically the observations with the largest variance, the PPML estimator gives equal weight to all observations.

A second approach uses a simulated method of moments (SMM) estimator, following Bernard et al. (2003), Chor (2009), Ramondo (2008), and Simonovska and Waugh (2014).²⁵ In our estimation, we search for parameter values that deliver predicted cross-border acquisition flows that match the first two moments of the actual data as closely as possible. We provide a more detailed description of this approach in the Internet Appendix.

We report results from cross-sectional regressions in Table III, featuring our preferred PPML estimator approach with instrumented regulatory variables. Differences in regulations are important determinants of cross-border bank acquisition flows.²⁶ The coefficients on each of the regulation indexes are positive and statistically reliable in all five cases, consistent with regulatory arbitrage. Cross-border bank acquisitions are more likely to involve acquirers from countries with more restrictions on bank activities, more capital requirements, more stringent supervision, better private monitoring, and better overall regulations relative to the targets. These results hold after controlling for additional factors that affect cross-border M&A activity, including target and acquirer country fixed effects.²⁷

²⁵ SMM has been used in the trade literature to truncate distributions of the Type 1 extreme value (EV) class (including Gumbel and Fréchet distributions) in order to accommodate zero values. Unlike the Type 1 EV distributions used by Chor (2009) and others, the statistical assumptions behind the PPML estimator have no difficulty predicting zeros. The Internet Appendix provides a detailed note comparing SMM and PPML approaches and explains our preference for PPML in our remaining analyses.

²⁶ The number of observations drops to a maximum of 2,599 (from 4,692) because of missing control variables and because some countries have missing information on instrumental variables.

²⁷ Our use of target and acquirer country fixed effects is in line with the way gravity models are estimated. Anderson and van Wincoop (2003) show the importance of controlling for multilateral trade resistance when estimating a gravity equation. This is done by using exporter and importer fixed effects in a cross-sectional test.

Table III
Cross-Sectional Determinants of Cross-Border Bank Acquisitions

The table shows estimates from cross-sectional regressions of majority cross-border bank acquisitions by country pairs. The dependent variable is the *cross-border ratio*—the total number of majority cross-border bank acquisitions between 1995 and 2012 in which the target is from country i and the acquirer is from country j ($i \neq j$), as a proportion of all domestic and cross-border bank acquisitions in target country i during the period. Independent variables are computed as differences (Δ) between acquirer country j and target country i . Detailed definitions of all variables are provided in Appendix B. To address endogeneity concerns, we instrument the regulatory variables using the government ownership of banks in 1970, the number of systemic banking crises in the 1970s and 1980s, and the percentage number of years since 1776 that each country has been independent. Specifically, in the first stage, we regress the differences in each of the regulatory variables on the differences in these instruments between acquirer and target countries and all of the controls used in the second stage. Columns (1) to (5) show results from regressions using Poisson pseudomaximum likelihood (PPML) estimation following Santos Silva and Tenreiro (2006). In columns (6) and (7), we present results from two-stage least squares (2SLS) regressions and a simulated method of moments (SMM) approach following Chor (2009), respectively, for the overall regulation variable. We include acquirer and target country fixed effects in all regressions and cluster standard errors by target country. We report pseudo- R^2 based on the differences in likelihood function values adjusted for the degrees of freedom for the PPML estimation, and the adjusted R^2 that account for degrees of freedom in the 2SLS regressions. The Hansen (Sargan) test of overidentifying restrictions assesses the joint null hypothesis that the instruments (IVs) are valid instruments, or uncorrelated with the error term, and that the excluded instruments are correctly excluded from the estimated equation. Under the null, the test statistic is distributed as chi-squared (χ^2) in the number of overidentifying restrictions. A rejection casts doubt on the validity of the instruments. For the efficient GMM estimator in the SMM approach, the Hansen's J -test statistic is the minimized value of the GMM criterion function. Under the assumption of conditional homoskedasticity, Hansen's J -statistic becomes Sargan's statistic. The J -statistic is consistent in the presence of heteroskedasticity and autocorrelation; Sargan's statistic is consistent in the presence of homoskedasticity and autocorrelation. For the 2SLS estimator, Sargan's statistic is typically calculated as the product of the number of observations and the R^2 from a regression of the IV residuals on the full set of instruments. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: Cross-Border Ratio _{<i>i,j</i>}						
	PPML – IV					2SLS	SMM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ restrictions on bank activities _{<i>j-i</i>}	3.196*** (3.15)						
Δ stringency of capital regulation _{<i>j-i</i>}		0.312* (1.94)					
Δ official supervisory power _{<i>j-i</i>}			0.383*** (3.16)				
Δ private monitoring _{<i>j-i</i>}				1.349*** (3.43)			
Δ regulation overall _{<i>j-i</i>}					1.092*** (3.43)	0.151** (2.18)	0.044* (1.92)
Δ governance index _{<i>j-i</i>}	3.949*** (2.75)	7.738*** (5.00)	8.367*** (5.15)	4.485*** (3.28)	7.092*** (4.84)	0.310* (-1.77)	0.165*** (3.14)
Δ log GDP per capita _{<i>j-i</i>}	6.469*** (4.09)	2.795*** (5.34)	2.653*** (4.36)	2.389*** (3.95)	2.472*** (4.02)	0.187 (1.61)	-0.059*** (-2.90)

(Continued)

Table III—Continued

	Dependent Variable: Cross-Border Ratio _{it,j}						
	PPML – IV					2SLS	SMM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ GDP growth _{j-i}	3.041*** (3.03)	0.341 (1.63)	0.435 (1.12)	-0.489 (-1.20)	0.172 (0.45)	0.003 (0.11)	-3.875*** (-3.16)
Δ bank credit _{j-i}	6.794*** (3.97)	2.927*** (4.91)	2.990*** (4.01)	0.767 (1.10)	2.363*** (3.46)	0.020 (1.21)	-0.022 (-0.44)
Δ bank concentration _{j-i}	16.250** (2.41)	-6.175*** (-3.51)	-3.705* (-1.92)	-4.296** (-2.20)	-3.246* (-1.66)	0.744 (1.59)	-0.411*** (-2.95)
Δ real exchange rate return _{j-i}	2.256*** (3.41)	0.061 (0.34)	0.700** (2.46)	-0.019 (-0.07)	0.510* (1.89)	-0.025** (-2.22)	0.100 (0.49)
Δ real stock market return _{j-i}	66.984*** (3.88)	27.071*** (4.28)	31.104*** (3.90)	24.613*** (3.38)	27.974*** (3.61)	0.600 (0.98)	-0.164 (-1.54)
Same language	1.134*** (4.55)	1.136*** (4.56)	1.134*** (4.55)	1.132*** (4.52)	1.132*** (4.52)	0.457*** (3.78)	0.176*** (2.74)
Colony	1.290*** (4.42)	1.292*** (4.43)	1.290*** (4.42)	1.295*** (4.40)	1.295*** (4.40)	0.939*** (2.94)	0.326* (1.80)
Contiguous	-0.072 (-0.24)	-0.104 (-0.35)	-0.072 (-0.24)	-0.071 (-0.24)	-0.071 (-0.24)	0.467* (1.82)	0.376*** (2.76)
Distance	-1.131*** (-9.03)	-1.136*** (-9.06)	-1.131*** (-9.03)	-1.131*** (-8.96)	-1.131*** (-8.96)	-0.090* (-1.86)	-0.034* (-1.69)
Bilateral trade	0.340 (0.32)	0.383 (0.36)	0.340 (0.32)	0.330 (0.31)	0.330 (0.31)	7.117*** (4.00)	2.385*** (2.78)
Δ growth opportunities _{j-i}	-3.485*** (-4.68)	-2.672*** (-4.72)	-2.916*** (-4.69)	-1.286** (-2.56)	-2.397*** (-4.34)	0.053 (1.53)	-0.936** (-2.00)
Target country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,599	2,599	2,599	2,550	2,550	2,550	2,550
Pseudo R ² (Adjusted R ²)	0.350	0.350	0.350	0.360	0.360	0.260	
Hansen's J-statistic (χ ²)							
p-value	0.317	0.317	0.318	0.371	0.982	0.120	0.063

The results are statistically and economically significant. Take, for instance, the coefficient on the *Official supervisory power* index in Model (3) (0.383 with a robust *t*-statistic of 3.16); a one- σ increase in the difference in that index (2.84, or the difference between the United States and Jordan) is associated with a 0.47% increase in the *Cross-border ratio*, which represents 42% of its σ .²⁸ Results are of similar magnitude for the other regulatory variables. The coefficient on the *Stringency of capital regulation* index in Model (2) (0.312, with a robust *t*-statistic of 1.94) implies that a one- σ increase in the difference in the index (1.77, which corresponds to the difference between the United Kingdom and Thailand) is associated with a 0.18% increase in the *Cross-border ratio*, or a still sizeable 16% of its σ . Consistent with the cross-border acquisitions literature, the results in Table III show that governance is positively correlated with cross-border bank acquisition flows. In addition, as in the trade literature,

²⁸ The coefficients in PPML should be interpreted as if the dependent variable is in logs. Thus, given the coefficient on the *Official supervisory power* index (0.383), a one- σ increase (2.84 units) is associated with a 2.97 times ($e^{0.383 \times 2.84}$) increase in the mean *Cross-border ratio* from 0.238% to 0.706%, or an increase of 42% of its σ (1.105%).

there are significantly more cross-border acquisitions between countries that share the same language, those that have a colonial link, and those that are geographically closer. Cross-border acquisition flows are also stronger between countries that share a border and those that have more intense bilateral trade, although the coefficients are not statistically reliable across specifications. In most specifications, cross-border bank acquisition flows are positively related to differences in GDP per capita, banking sector size, stock market returns, and real exchange rates. Acquirers appear to pursue targets in more concentrated banking sectors. The Hansen J -statistic (χ^2) test cannot reject the joint null that the instruments are valid.

In Models (6) and (7) of Table III, we show results for the overall regulation index using alternate estimation methods. In Model (6), we show results from 2SLS regressions, while Model (7) displays equivalent results using SMM. Consistent with our findings using PPML, the results continue to show that differences in bank regulation are positively associated with cross-border acquisition flows. Assessing the magnitude of the results from 2SLS estimation, the coefficient on the differences in the overall regulation index (0.151) in Model (6) implies that a one- σ increase in the difference (1.08) is associated with a 0.16% increase in the *Cross-border ratio*, or 15% of its σ .²⁹ The coefficient on the difference in the overall regulation index using SMM is smaller in magnitude, though still reliably different from zero. The coefficient on the overall regulation index built from principal component analysis (0.044, robust t -statistic of 1.92) implies only a 0.05% increase in the *Cross-border ratio*, which is smaller relative to its average value of 0.24%.

Comparing the control variables across the different estimation methods yields several interesting observations. The positive coefficients on *Governance index*, *Same language*, *Colonial link*, and *Log GDP per capita* are robust across all three estimation methods, as is the negative coefficient on *Distance*. Furthermore, *Bilateral trade* and *Contiguous* become significant in both the SMM and 2SLS estimations. For the SMM estimation, the positive coefficients on *Bank credit* and *Real stock market return* disappear, but the difference in *Log GDP per capita* now takes a significantly negative coefficient (acquirers pursue targets in more developed countries). In only one specification in Table III (Model (7) for SMM) can we reject the overidentifying restrictions associated with our set of instrumental variables using Hansen's J -statistic. We report the full set of results from 2SLS and SMM regressions (using all five regulatory variables) in the Internet Appendix.

Overall, our results in this section document a positive correlation between differences in regulation and the flow of majority cross-border bank acquisitions during our sample period. This approach cumulates flows by country pairs across the entire sample period and thus suppresses any time-series variation.

²⁹ The coefficient on the overall regulation index (0.151) implies that a one- SD increase (1.08 units, from Argentina to the United States, for example) is associated with a 0.163% (0.151×1.08) increase in the mean *Cross-border ratio* (0.24%), or an increase of 15% of its SD (1.105%).

We next examine how differences in regulation explain annual cross-border flows using panel analysis.

C. Panel Data Analysis of Determinants of Cross-Border Bank Acquisitions

To further examine how differences in regulations across countries affect the volume and frequency of cross-border bank acquisitions, we run the following panel regression:

$$CB_{j,i,t} = \exp[\alpha + \beta \Delta REG_{j-i,t} + \gamma X_{j-i,t} + \delta_i + \vartheta_j + \tau_t] + \varepsilon_{j-i,t}, \quad (4)$$

where $CB_{j,i,t}$ is the total number of majority cross-border bank acquisitions in year t in which the target is from country i and the acquirer is from country j ($i \neq j$), scaled by the total number of domestic and cross-border bank acquisitions in target country i in year t (if a target country does not have any cross-border bank acquisition activity in a given year, we exclude that target country-year), $\Delta REG_{j-i,t}$ is a vector of variables measured as differences between the acquirer and target countries that includes our five indices of bank regulation that are now allowed to vary over the sample period, and $X_{j-i,t}$ is a vector of controls measured as differences between the acquirer and target countries that includes *Governance index*, *Log of GDP per capita*, *GDP growth*, *Bank concentration*, *Bank credit*, *Real exchange rate return* and *Real stock market return*, *Bilateral trade*, *Growth opportunities*, *Distance*, *Same language*, *Contiguous*, and *Colony*. Note that a number of these control variables are time-varying across years.

We include target and acquirer country fixed effects (δ_i, ϑ_j) in all regressions to control for other time-invariant country characteristics and year fixed effects (τ_t) to capture unobservable global market conditions. To address endogeneity concerns, as in Table III, we use instrumental variables for the differences (always defined as acquirer less target) in our bank regulatory measures. We report the first-stage regression results in the Internet Appendix. Diagnostic evidence in Panel A of Table IV shows that our estimates are unlikely to be biased due to weak instruments. The test of overidentifying restrictions (Hansen's J -statistic) cannot reject the null hypothesis that the instruments are valid except in Model (2), which uses the *Stringency of capital regulation* index.³⁰

³⁰ The Internet Appendix shows results from first-stage regressions. Differences in government ownership in the banking system as of 1970 (from La Porta, Lopez-de-Silanes, and Shleifer (2002)) between the acquirer and target countries are associated with positive differences in bank activities restrictions, *Official supervisory power*, and overall regulatory quality, but negative differences in *Stringency of capital regulation* and *Private monitoring*. Differences in the number of banking crises experienced in the 1970s and 1980s (from Laeven and Valencia (2012)) between the acquirer and target countries lead to positive differences in *Stringency of capital regulation* and *Private monitoring*, but negative differences in bank activities restrictions, *Official supervisory power*, and overall regulatory quality. Finally, differences in the proportion of years since 1776 that a country has been independent between acquirer and target countries lead to negative gaps in all regulatory quality variables. These results suggest that countries that have been independent longer tend to have less stringent regulations.

Table IV
Panel Regression Analysis of the Determinants of Cross-Border Bank Acquisitions

The table shows estimates from panel regressions of majority cross-border bank acquisitions by country-pair-year using the PPML estimator following Santos Silva and Tenreyro (2006). The dependent variable is the annual *Cross-border ratio* for each country pair—the total number of majority cross-border bank acquisitions in year t in which the target is from country i and the acquirer is from country j ($i \neq j$), as a proportion of all domestic and cross-border bank acquisitions in target country i in year t . If a target country does not have any cross-border bank acquisition activity in a given year, we exclude that country for that year. Independent variables are computed as differences between acquirer country j and target country i . See Appendix B for a description of the four indices of regulatory quality from the revised database from Barth, Caprio, and Levine (2013) and for a description of the independent variables. See notes to Table III on Hansen's J -statistic. In Panel A, we report results for the full sample, while Panel B shows results when we exclude all U.S. targets. To conserve space, we do not report the coefficients on the control variables in Panel B. We include acquirer country, target country, and year fixed effects (not shown) and cluster standard errors by target country. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Panel A: PPML Estimation with Instrumental Variables					
	Dependent Variable: Cross-Border Ratio $_{i,j}$				
	(1)	(2)	(3)	(4)	(5)
Δ restrictions on bank activities $_{j-i}$	0.544** (2.24)				
Δ stringency of capital regulation $_{j-i}$		1.240** (2.49)			
Δ official supervisory power $_{j-i}$			0.374* (1.95)		
Δ private monitoring $_{j-i}$				4.452*** (2.60)	
Δ regulation overall $_{j-i}$					4.640* (1.76)
Δ governance index $_{j-i}$	0.081 (0.17)	1.321* (1.65)	0.310 (0.66)	3.788** (2.15)	1.655 (1.08)
Δ log GDP per capita $_{j-i}$	1.307*** (3.28)	1.663* (1.85)	0.696** (2.28)	1.149 (1.20)	0.084 (0.06)
Δ GDP growth $_{j-i}$	-4.962* (-1.81)	-7.505** (-2.09)	-3.508 (-1.47)	6.573 (1.21)	-2.548 (-0.61)
Δ bank credit $_{j-i}$	-0.121 (-1.38)	-0.088 (-0.51)	-0.222** (-1.99)	-0.400 (-1.09)	-0.346* (-1.92)
Δ bank concentration $_{j-i}$	0.009 (1.29)	0.014 (1.28)	0.004 (0.73)	0.025* (1.92)	0.026* (1.92)
Δ real exchange rate return $_{j-i}$	-0.280 (-0.72)	-0.431 (-0.77)	0.070 (0.21)	-0.354 (-0.58)	0.229 (0.32)
Δ real stock market return $_{j-i}$	0.245 (0.74)	0.379 (0.89)	0.263 (1.02)	0.380 (0.74)	0.318 (0.93)
Same language	1.011*** (3.98)	1.073*** (3.41)	1.058*** (4.33)	1.052*** (2.87)	1.291*** (3.83)
Colony	0.693** (1.98)	0.452 (1.09)	0.738** (2.23)	1.034** (2.00)	0.675* (1.70)

(Continued)

Table IV—Continued

Panel A: PPML Estimation with Instrumental Variables					
Contiguous	0.391 (1.18)	0.313 (0.74)	0.350 (1.09)	0.506 (1.09)	0.549 (1.11)
Distance	−1.016*** (−7.19)	−0.899*** (−6.12)	−0.986*** (−7.48)	−1.285*** (−4.91)	−1.035*** (−4.09)
Bilateral trade	−1.096 (−1.09)	−0.453 (−0.36)	−0.680 (−0.71)	−2.075 (−1.46)	0.004 (0.00)
Δ growth opportunities $j-i$	1.182 (0.58)	1.936 (0.72)	−1.865 (−0.87)	−0.140 (−0.04)	−0.597 (−0.18)
Target country f.e.	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes
Observations	23,732	22,842	24,332	23,253	22,651
Pseudo R^2	0.053	0.018	0.048	0.017	0.017
First-stage F -statistic	562.14	64.23	83.19	482.97	81.78
p -value	0.000	0.000	0.000	0.000	0.000
Hansen's J -statistic (χ^2)	1.59	7.52	2.51	2.93	0.07
p -value	0.45	0.02	0.29	0.23	0.79
Panel B: PPML Estimation with Instrumental Variables—Excluding U.S. Targets					
	(1)	(2)	(3)	(4)	(5)
Δ restrictions on bank activities $j-i$	0.544** (2.57)				
Δ stringency of capital regulation $j-i$		0.736*** (3.16)			
Δ official supervisory power $j-i$			0.225** (2.31)		
Δ private monitoring $j-i$				1.657** (2.51)	
Δ regulation overall $j-i$					4.945** (2.32)
Controls	Yes	Yes	Yes	Yes	Yes
Target country f.e.	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes
Observations	22,004	21,278	22,478	21,474	20,998
Pseudo R^2	0.054	0.031	0.039	0.040	0.039
Hansen J -statistic (χ^2)	12.36	22.03	12.94	4.85	15.98
p -value	0.00	0.00	0.00	0.09	0.00

We present results from PPML estimation of equation (4) in Table IV. We find that the volume of cross-border bank acquisition activity between two countries is related in a statistically and economically important way to differences in the quality of bank regulations. Acquirers come from countries with more activities restrictions, stricter capital requirements, stronger supervision, stronger private monitoring, and stronger overall regulatory environments. For example, the coefficient on the difference in *Restrictions on bank activities* in Model (1) is 0.544 (robust t -statistic of 2.24) and that on *Stringency of capital*

regulation in Model (2) is 1.24 (robust t -statistic of 2.49). Coefficients on the remaining regulatory variables are positive and reliably different from zero. The results are economically large. For example, a one- σ increase in the difference in activities restrictions for a given country pair (2.96, which roughly corresponds to the difference between the United States and Qatar) is associated with a 1.04% increase in the *Cross-border ratio*, or 37% of its σ .³¹

Regulatory arbitrage appears to be a motivating factor in cross-border bank acquisitions, even in a dynamic panel setting and after accounting for many additional factors that may affect cross-border acquisition activity. Consistent with prior findings in the corporate literature on cross-border mergers (Rossi and Volpin (2004), Bris and Cabolis (2008), Starks and Wei (2013)), governance plays a role in cross-border bank acquisition flows; there is more activity in which acquirers come from countries with better governance. However, in certain specifications—namely, Models (1), (3), and (5)—the broader measure of governance loses much of its explanatory power after controlling for differences in bank regulation. This suggests that differences in regulation are more important than differences in broad governance measures for cross-border bank acquisitions. We also find some evidence that acquirers in cross-border bank acquisitions pursue targets from less developed countries with faster growth and larger banking sectors. Finally, consistent with our cross-sectional findings on cross-border acquisitions and in line with findings in the trade literature, we find that cross-border bank acquisition activity is more pronounced between countries that are geographically more proximate and between countries that share the same language or a colonial link. While we only present results from PPML estimations here, we report results from 2SLS regressions in the Internet Appendix. We obtain similar results. Coefficients on all regulatory variables are positive and statistically significant (at the 10% level or better). The control variables retain similar signs for the most part. Coefficients on *Bilateral trade* and *Contiguous* show up reliably significant and positive in all specifications.

While our sample of bank acquisitions has a large number of countries, the United States is by far the largest target market and could weigh heavily on our inferences. We explore this possibility in Panel B of Table IV, where we present results from PPML estimation of equation (4) excluding all U.S. targets. To conserve space, we do not show the coefficients on the control variables.³² In each of the five specifications, differences in bank regulation are important factors in cross-border bank acquisition flows. The coefficients on all bank regulation variables are positive and statistically significant, in line with Panel A. The magnitude of the coefficients is similar, although the coefficients on *Stringency of capital regulation*, *Official supervisory power*, and *Private monitoring* are

³¹ The average annual *Cross-border ratio* across all country-pair-years is 0.26% (Appendix C). Coefficients in PPML should be interpreted as if the dependent variable is in logs. Thus, given the coefficient on activities restrictions in Panel A of Table IV (0.544), a one- SD increase (2.96 units) is associated with a five times ($e^{0.544 \times 2.96}$) increase in the mean *Cross-border ratio* from 0.26% to 1.30%, or an increase of 37% of its σ (2.78).

³² We show the full set of results in the Internet Appendix.

somewhat smaller, while the coefficient on *Regulation overall* is larger when we remove U.S. targets from the analysis.

Our findings on cross-border bank acquisition flows accord well with the main findings in Houston, Lin, and Ma (2012) that bank flows, which in their study are measured by changes in total foreign claims from a given source country to another recipient country based on data from the BIS, are more pronounced from countries with stronger bank regulations to those with weaker ones. While Houston, Lin, and Ma (2012) conclude that their evidence points to a destructive race-to-the-bottom form of regulatory arbitrage, we do not render a verdict on regulatory arbitrage and cross-border bank acquisition patterns without further analysis at the deal level.

III. Regulatory Arbitrage and Stock Price Reactions to Acquisition Announcements

We next examine how differences in banking regulations influence stock price reactions to cross-border bank acquisition announcements. To capture stock price reactions to bank acquisition announcements, we calculate cumulative abnormal returns (CAR) around the announcement date for both targets and acquirers using a market model, where we use the world market index to proxy for the market return

$$R_{ijkt} = \alpha_{ijk} + \beta_{ijk}^w R_{wt} + \varepsilon_{ijkt} \quad t = -260, \dots, -21. \quad (5)$$

In (5), R_{ijkt} is the day t stock return for bank k (target or acquirer) involved in a deal from acquirer country j and target country i , R_{wt} is the world market index, and ε_{ijkt} is the error term. We then cumulate the abnormal returns over event windows $(t_1, t_2) = (-20, +20)$, $(-1, +1)$, and $(-2, +2)$. The CAR is computed as follows:

$$CAR_{ijk}^{(t_1, t_2)} = \sum_{t=t_1}^{t=t_2} (\hat{\varepsilon}_{ijkt}). \quad (6)$$

Estimating abnormal returns using a world market index facilitates comparison of abnormal returns across countries, but we also estimate CARs using national stock market indexes. While we cumulate abnormal returns over various event windows, our regression tests focus primarily on abnormal returns cumulated over the five days $(-2, +2)$ surrounding the announcement date. Results are similar when we use abnormal returns cumulated over the three-day window $(-1, +1)$ as well as longer windows.³³

Table V reports descriptive statistics for targets and acquirers. Panel A compares abnormal returns for targets and acquirers in domestic and cross-border acquisitions. The results show that targets in domestic (cross-border) bank acquisitions experience a 14.9% (5.8%) CAR over the five-day period surrounding

³³ The Internet Appendix reports results using abnormal returns cumulated over the three days surrounding the announcement as well as over longer windows: $(-20, +20)$ and $(-100, 0)$.

Table V
Cumulative Abnormal Returns for Targets and Acquirers

The table shows descriptive statistics of cumulative abnormal returns (CARs) for targets and acquirers in completed domestic and cross-border majority acquisitions announced between 1995 and 2012. To compute abnormal returns, we estimate the following market model for each target (acquirer):

$$R_{i,jkt} = \alpha_{ijk} + \beta_{ijk}^w R_{wkt} + \varepsilon_{ijkt}, t = -260, \dots, -21,$$

where $R_{i,jkt}$ is the daily stock return for target (acquirer) k involving a deal with an acquirer from country j and a target from country i , R_{wkt} is the world market index, and ε_{ijkt} is the excess return for each bank. All returns for international banks are obtained from DataStream; the returns for U.S. institutions are obtained from CRSP. We report CARs over three event windows: $(-1, +1)$, $(-2, +2)$, and $(-20, +20)$. We winsorize all CARs at the top and bottom 1% of the distribution. Panel A shows descriptive statistics for CARs for targets (acquirers) in domestic and cross-border deals. Panel B shows descriptive statistics of the acquirers and targets in majority bank acquisitions. Financial information on targets and acquirers comes from WorldScope and is complemented with data from Bloomberg. Data are as of the year-end prior to the announcement of the deal. Total assets are in US\$ million. Return on assets (ROA) is net income plus interest expense divided by total assets. Z-score is $(\text{ROA} + \text{equity/assets})/\sigma(\text{ROA})$, where the σ of ROA, $\sigma(\text{ROA})$, is estimated as a three-year moving average. We report the log of Z-score, following the literature (e.g., Laeven and Levine (2009)). NPL-to-G/L is total nonperforming loans (past due 90 days or more) as a share of gross loans. The last column of each panel reports p -values from tests of differences in mean (t -tests) and median (Wilcoxon matched-pairs signed rank test). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Cross-Border Acquisitions				Panel A: Cross-Border and Domestic Acquisitions				Test of Difference	
					Domestic Acquisitions				<i>t</i> -statistic	Wilcoxon Z-statistic
	Mean	Median	σ	<i>N</i>	Mean	Median	σ	<i>N</i>		
Targets:										
CAR (-2, +2)	0.058**	0.026	0.218	86	0.149***	0.125	0.217	994	(0.000)	(0.000)
CAR (-20, +20)	0.082***	0.033	0.234	86	0.188***	0.171	0.254	990	(0.000)	(0.000)
CAR (-1, +1)	0.053**	0.019	0.209	86	0.145***	0.118	0.210	994	(0.000)	(0.000)
Acquirers:										
CAR (-2, +2)	-0.003*	-0.003	0.037	676	-0.001	-0.002	0.052	3,307	(0.199)	(0.396)
CAR (-20, +20)	-0.004	-0.004	0.092	672	-0.003	-0.002	0.117	3,281	(0.852)	(0.731)
CAR (-1, +1)	-0.002*	-0.002	0.030	676	-0.002***	-0.002	0.042	3,307	(0.926)	(0.776)

(Continued)

Table V—Continued

Panel B: Description of Sample of Majority Acquisition Firms										
Cross-Border Acquisitions					Domestic Acquisitions					Test of Difference
	Mean	Median	σ	N	Mean	Median	σ	N	t -statistic	Wilcoxon Z -statistic
Targets:										
Total assets	15,803	2,060	40,333	162	4,570	75	24,161	1,919	(0.000)	(0.000)
ROA	2.05%	1.73%	2.87%	135	3.34%	3.79%	2.41%	852	(0.000)	(0.000)
Z-score	2.68	2.85	1.28	81	2.93	2.92	1.08	633	(0.680)	(0.956)
Acquirers:										
Total assets	362,290	169,713	445,103	762	45,349	75	173,537	4,280	(0.000)	(0.000)
ROA	1.52%	1.25%	1.25%	657	3.18%	3.04%	2.07%	3,524	(0.000)	(0.000)
Z-score	3.05	2.99	0.90	632	3.11	3.03	0.94	2,739	(0.466)	(0.167)
NPL-to-GL	1.92%	1.42%	2.17%	528	0.68%	0.83%	1.98%	3,599	(0.000)	(0.000)

the announcement. Targets in domestic deals experience significantly larger abnormal returns than targets in cross-border deals. The difference is economically large, at about 9.1% and the mean and median reactions are significantly different (p -values in last two columns). These figures are well within the ranges in Campa and Hernando (2006), which also establish smaller reactions for cross-border targets. The results using alternate windows $(-1, +1)$ and $(-20, +20)$ are similar, with larger CARs for targets in domestic acquisitions. Consistent with Amihud, DeLong, and Saunders (2002), the results in Panel A uncover negative CARs for acquirers in both domestic and cross-border deals. The magnitudes of the acquirer returns are small and not always statistically significantly different from zero.

Panel B of Table V shows descriptive statistics for acquirers and targets in domestic and cross-border bank acquisitions. Sample size varies depending on which variable is used. Acquirers tend to be larger but less profitable (as measured by return on assets, ROA) than targets, not only in domestic but especially in cross-border deals. Acquirers in cross-border deals are larger, less profitable, and riskier than their counterparts in domestic acquisitions (higher level of nonperforming loans). Targets in cross-border deals are also larger, less profitable, and riskier than their counterparts in domestic deals.

Given that our objective is to examine the motives behind regulatory arbitrage in cross-border bank acquisitions, we need to distinguish share price reactions by controlling for additional firm-, deal-specific and country-level characteristics, which we explore next. While we focus on determinants of target CARs, we also examine acquirer CARs and aggregate CARs combining both targets and acquirers.³⁴

Turning to the determinants of share price reactions to cross-border bank acquisition announcements, we focus on how differences in regulation affect value creation around such deals. We first analyze the determinants of target CARs around the announcement. If regulatory arbitrage is driven by a costly race to the bottom, whereby acquirers escape tough regulatory regimes by acquiring targets in weaker regimes to pursue value-destroying activities, we expect less positive (and possibly negative) stock price reactions from a target's shareholders around the deal announcement, all else—including terms of the deal—being equal. In contrast, more positive reactions can arise if acquirers use regulatory arbitrage to expand their business and diversify their activities, especially when they are constrained by costly restrictions in their home country. A target bank's shareholders may welcome deals involving acquirer banks from tougher regimes. They may gain from receiving a capital infusion from a

³⁴ Our goal is to pin down the value gains acquirers and targets realize from acquisitions in different regulatory environments. An important factor in the target bank's share price reaction is the bid premium relative to the preannouncement closing price offered by the acquirer. Unfortunately, we are able to obtain bid premiums (bid price relative to close one day prior to announcement) for only 41 cross-border and 1,009 domestic acquisitions over our sample period. Consistent with the magnitude of the target CARs, we find that the average cross-border (domestic) premium is 17.8% (31.6%). We show results using the bid premium as a dependent variable in our cross-sectional regression tests in the Internet Appendix.

bank that brings increased managerial discipline or better risk-management techniques, that is, from being bonded to the tougher regime by means of the cross-border acquisition.

To examine these competing views on regulatory arbitrage, we run the following regressions:

$$CAR_{ijkt} = \alpha_{ijk} + REG_{j-i,t} + \varphi C_{j-i,t} + \delta B_{j-i,t} + \theta_i + \vartheta_j + \tau_t + \varepsilon_{ijkt}, \quad (7)$$

where CAR_{ijkt} is the year t CAR of bank k (target, acquirer, aggregate) involved in a deal from acquirer country j and target country i . As in Section II, $REG_{j-i,t}$ is a vector of differences in the proxy variables for the quality of bank regulation between acquirer country j and target country i in year t ; to mitigate endogeneity concerns, we present results in which we instrument our regulatory variables.³⁵ Next, $C_{j-i,t}$ is a vector of country characteristics, measured as differences between acquirer country j and target country i in year t , which includes *Governance index*, *Log of GDP per capita*, *GDP growth*, *Bank concentration*, *Bank credit*, *Real exchange rate return* and *Real stock market return*, *Bilateral trade*, *Growth opportunities*, *Distance*, *Same language*, *Contiguous*, and *Colony*. The vector $B_{j-i,t}$ contains bank-level controls, measured as differences between the acquirer and target banks, which includes total assets (in logs) in most regressions. Large differences in size between a target and acquirer may significantly affect the outcome of the deal, and as such the abnormal returns obtained from such deals.³⁶ We include year dummy variables (τ_t) in all regressions. In addition, we include acquirer and target country fixed effects to control for other plausible time-invariant country characteristics that may affect stock price reactions to the deals.

Results from estimation of equation (7) are reported in Table VI. Panel A presents results for target CARs over the $(-2, +2)$ window. The results highlight the importance of differences in the quality of bank regulations between the acquirer and the target countries. In all models, target banks' CARs are higher when the acquirer comes from a country with more restrictions on bank activities, stricter capital requirements, and stronger private monitoring. While differences in *Official supervisory power* are also positively correlated with target CARs, the result is not statistically significant. Most importantly, the coefficient on our overall measure of regulatory quality is positive and statistically significant. Furthermore, the impact is economically sizeable. The estimates from Model (1) show that a one- σ increase in the difference in *Restrictions on bank activities* (1.48 points) is associated with 0.14% higher CARs than the unconditional average 8.72% market reaction. From Model (2), we

³⁵ As before, our instruments include government ownership of the banks in a country as of 1970 from La Porta, Lopez-de-Silanes, and Shleifer (2002), the number of systemic banking crises in country i during the 1970s and 1980s from Laeven and Valencia (2012), and the percentage of years since 1776 that a country has been independent from Beck, Demirgüç-Kunt, and Levine (2006).

³⁶ Deal completion rates among our cross-border and domestic bank acquisitions are both similarly high. Table I indicates that 7,257 of the 7,297 announced domestic deals were completed, while 901 of 916 cross-border deals were completed.

Table VI
Regressions of Cumulative Abnormal Returns

The table shows results from 2SLS regressions of CARs for targets and acquirers in domestic and cross-border bank acquisitions over the period 1995 to 2012. The dependent variable represents CAR for targets (targets and acquirers) over the period $t = (-2, +2)$ around the announcement day, unless otherwise noted. See notes to Table V for a description of the estimation of CARs. All returns for international banks are obtained from DataStream, while the returns for U.S. institutions are obtained from CRSP. Independent variables are computed as differences between acquirer country j and target country i . See Appendix B for a description of the indices of regulatory quality from the revised database from Barth, Caprio, and Levine (2013), as well as the additional control variables. To address endogeneity concerns, we instrument the regulatory variables using the government ownership of banks in 1970, the number of systemic banking crises in the 1970s and 1980s, and the percentage number of years since 1776 that each country has been independent. In Panel A, we show results from 2SLS regressions using target CARs as the dependent variable. In Panel B, we present additional robustness tests. In columns (1) and (2) of Panel B, we show results for acquirer and aggregate CARs (market-value-weighted average of acquirer and target CARs), respectively. In column (3), we show results for target CARs excluding deals involving U.S. targets. Finally, in columns (4) and (5), we show results using longer windows for target CARs. To conserve space, we do not show the coefficients on the control variables in Panel B. We include acquirer country, target country, and year fixed effects (not shown) in all regressions and cluster standard errors by target country. Additional details on regression diagnostics are furnished in the caption for Table III. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Panel A: Target CARs (-2, +2)					
	(1)	(2)	(3)	(4)	(5)
Δ restrictions on bank activities $j-i$	0.095*** (4.63)				
Δ stringency of capital regulation $j-i$		0.540*** (7.67)			
Δ official supervisory power $j-i$			0.035 (0.92)		
Δ private monitoring $j-i$				0.025*** (2.75)	
Δ regulation overall $j-i$					0.098*** (2.65)
Δ governance index $j-i$	-0.238** (-2.55)	-0.197 (-1.05)	-0.039 (-0.34)	0.327* (1.95)	1.909*** (13.41)
Δ log GDP per capita $j-i$	-0.433*** (-3.49)	0.111 (0.31)	-0.200*** (-3.65)	-0.382*** (-5.04)	-0.877*** (-15.47)
Δ GDP growth $j-i$	-1.255* (-1.72)	-7.600*** (-6.10)	-1.878* (-1.91)	-2.864*** (-2.96)	-3.047*** (-9.46)
Δ bank credit $j-i$	-0.001 (-0.47)	-0.006*** (-3.10)	-0.001 (-0.73)	-0.000 (-0.23)	-0.008*** (-17.14)
Δ bank concentration $j-i$	0.013*** (11.43)	0.004 (1.34)	0.009*** (5.79)	0.012*** (8.75)	0.019*** (27.03)
Δ real exchange rate return $j-i$	-0.727** (-2.42)	3.140*** (6.48)	-0.405 (-1.31)	-0.258 (-0.66)	-0.058 (-0.35)
Δ real stock market return $j-i$	-0.014 (-0.22)	-0.828*** (-3.99)	-0.092 (-0.92)	0.059 (0.56)	-0.213*** (-4.42)
Same language	-0.216** (-2.43)	-0.284 (-1.40)	0.010 (0.11)	-0.235** (-2.29)	-2.041*** (-25.15)

(Continued)

Table VI—Continued

Panel A: Target CARs (−2, +2)					
	(1)	(2)	(3)	(4)	(5)
Colony	−0.287*** (−3.39)	0.050 (0.97)	−0.328 (−1.52)	−0.070 (−0.63)	−0.009 (−0.07)
Contiguous	−2.265*** (−12.85)	5.146*** (6.28)	−1.758*** (−6.89)	0.737*** (3.17)	4.733*** (16.82)
Distance	0.011 (0.21)	0.314*** (6.46)	−0.064 (−1.01)	−0.014 (−0.63)	1.015*** (25.72)
Bilateral trade	2.226*** (4.61)	−14.070*** (−8.27)	2.621*** (6.06)	1.662 (1.34)	−11.995*** (−20.62)
Δ growth opportunities $j-i$	1.114*** (2.99)	1.267*** (3.22)	0.131 (0.43)	0.010 (0.03)	−0.269 (−1.50)
Δ size $j-i$	0.003*** (10.32)	0.002*** (9.22)	0.003*** (9.69)	0.003*** (10.42)	0.002*** (8.88)
Target country f.e.	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes
Observations	867	866	867	867	866
Adjusted R^2	0.210	0.207	0.209	0.212	0.215
Partial R^2	0.90	0.07	0.18	0.57	0.29
First-stage F -statistic	101.64	34.11	61.45	78.47	44.15
p -value	0.00	0.00	0.00	0.00	0.00
Sargan χ^2 statistic	3.88	0.68	0.01	2.54	0.07
p -value	0.14	0.41	0.92	0.11	0.79
Panel B: Alternate CARs					
	Acquirer CARs (1)	Aggregate CARs (2)	Target CARs No U.S. Targets (3)	Target CARs (−20, +20) (4)	Target CARs (−100, +100) (5)
Δ regulation overall $j-i$	0091* (182)	0.038* (1.68)	0.170*** (2.88)	0.092*** (3.15)	0.617*** (2.78)
Controls	Yes	Yes	Yes	Yes	Yes
Target country f.e.	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes
Observations	1,256	623	163	865	865
Adjusted R^2	0101	0.214	0.545	0.038	0.017
Partial R^2	002	0.19	0.09	0.28	0.15
First-stage F -statistic	252	90.16	34.49	44.33	40.95
p -value	009	0.00	0.00	0.00	0.00
Sargan χ^2 statistic	014	1.84	1.61	0.02	0.00
p -value	070	0.40	0.45	0.89	0.97

uncover an even larger difference: a one- σ increase in the stringency of capital regulation index (1.19 points) is associated with a 0.64% higher reaction, on average. Overall, target bank shareholders reward cross-border acquisitions in which the acquirers come from better regulatory environments in terms

of tougher restrictions on bank activities, stricter capital requirements, and stronger private monitoring. These results suggest that target shareholders view this influence of bank regulation positively, which is inconsistent with the harmful view of regulatory arbitrage in cross-border bank acquisitions.

In addition to the importance of differences in bank regulation, the results show that target CARs are significantly higher when targets are from relatively better developed, faster growing countries and with less concentrated banking sectors than their acquirer countries. In contrast, there is some evidence of a larger positive target reaction to deals involving larger acquirers, acquirers from countries with smaller banking sectors, and acquirers from countries with more growth opportunities. The evidence on the impact of governance after controlling for the quality of banking regulation is mixed. In Model (1), CARs are significantly lower when acquirers come from countries with better governance, but the opposite is the case in Models (4) and (5), where the coefficient is positive and statistically significant (as for the cross-border deals in Bris and Cabolis (2008)).

Regression diagnostics in Table VI show that our specifications are reasonable. The adjusted R^2 averages around 20%. The first-stage F -statistics consistently support the relevance of the instrumental variables we have selected and we are unable to reject the joint null of their validity using the Sargan test of overidentifying restrictions in any of the five models.

So far, we have shown that target shareholders reward cross-border deals that are in line with regulatory arbitrage, which suggests that such deals do not reflect the destructive race-to-the bottom form of regulatory arbitrage. In Panel B of Table VI, we repeat results for cross-sectional regressions of acquirer and aggregate acquirer-target CARs in Models (1) and (2), respectively. To conserve space, we repeat results only for the overall regulation index and we do not report the coefficients on the control variables. To obtain a measure of aggregate CARs, we weight the acquirer and target banks' respective CARs using their relative market values as of the time of the deal. In Models (1) and (2), we report results for acquirer and aggregate CARs, respectively. We continue to find a positive and significant relation between differences in regulatory quality, though of much smaller size. Given the large difference in size between acquirer and target banks in these cross-border deals (see Panel B of Table V), it is somewhat surprising that target CARs drive the results. The economic importance of the transaction for target banks thus appears to outweigh the neutral reaction for most acquirers in spite of the targets' smaller relative size.

In Model (3) of Panel B, we report results of target CARs excluding targets from the United States. The results continue to show a more positive target share price reaction for deals involving acquirers from stronger regulatory regimes. The magnitude of the coefficient on differences in overall regulatory quality is actually larger than when U.S. targets are included. Finally, in Models (4) and (5) of Panel B, we expand our CAR windows in an attempt to capture possible information leakage prior to the announcement of the acquisitions. The results continue to show a positive and significant target share price

reaction for deals involving acquirers from countries with a better regulatory environment.

To evaluate the robustness of our results, we run various additional tests, which are available in the Internet Appendix. Specifically, we replicate our results using the raw regulatory variables instead of the instrumented variables; we exclude targets and acquirers from the United States, which comprise the bulk of our target sample; we exclude pure cash deals to address concerns that the favorable reaction of target shareholders to cross-border acquisitions is mechanically due to the bid premium paid; we incorporate additional controls for stock-financed deals; and we evaluate the reactions over a shorter three-day event window. Excluding pure cash deals reduces our sample to about 589 deals from the 867 featured in Table VI. The economic magnitudes of the reactions are smaller, but we still observe reliably positive target CARs for four of the five regulatory indexes. The three-day target CAR results are weaker and the magnitudes of the reactions are smaller, which is not surprising given the results in Table V, but the aggregate CARs are still reliably positive. Finally, we perform regressions using alternate CAR windows and find results consistent with those reported in Table VI.³⁷ Our results consistently show that target CARs are higher when acquirers come from countries with better regulatory quality.

IV. The Impact of Merger Control Policies on Regulatory Arbitrage

A. Changes in Merger Control Policies and Cross-Border Bank Acquisition Flows

Our results thus far show that cross-border bank acquisition flows tend to involve acquirers from countries with tougher regulatory regimes than their targets. But the completion of cross-border bank acquisitions depends not only on the decisions of the institutions that are party to the deal; they typically also require various types of regulatory approval. While we attempt to control for additional factors that may have an impact on cross-border bank acquisition flows (e.g., we include country and year fixed effects in all our regressions), we are potentially ignoring an important determinant of such flows, namely, merger control policies.³⁸ Merger control policies, which are a key component of competition policies in general, ultimately determine which mergers and acquisitions will take place. Importantly, over the last few decades, several countries have introduced legislation aimed to strengthen merger control policies (see, among others, Carletti (2008), Carletti and Vives (2009)).

Given the importance of merger control policies, in this section, we examine how changes in merger control affect our results. The main difficulty we face is to find adequate measures that capture time-varying stringency of merger control, which is especially challenging considering our large sample of countries and our sample period that spans a few decades. To date, few studies

³⁷ Specifically, we report results using all five measures of regulatory quality for regressions of target CARs using the following windows: (−20, +20) and (−100, 0).

³⁸ We thank an anonymous referee for raising this important issue.

examine time-varying changes in merger control policies and their effect on cross-border acquisitions, and only recently has there been an effort to construct indices that capture such changes in merger control (e.g., Buccirosi, et al. (2011, 2013)). Measuring changes in merger control in the banking sector is further complicated by the fact that regulators from both acquirer and target countries may be involved. There exist specific control regulations for the financial sector in many countries (e.g., Carletti, Hartmann, and Ongena (2013)), and large cross-border bank acquisitions may involve transnational authorities (e.g., Duso, Neven, and Röller (2007)). In addition, political factors may also influence the enforcement of such controls (Duso, Gugler, and Yurtoglu (2011)). Köhler (2009), for example, shows that implicit government barriers (as measured by a lack of transparency in the regulatory process) are an obstacle to consolidation in the EU banking sector.

Acknowledging the difficulties inherent in adequately measuring merger control, we explore several ways to control for merger control in our analysis. First, we construct two measures using the fraction of failed merger and acquisition deals. To construct these measures, we obtain data from Thomson's SDC database on failed deals among all deals including nonfinancial firms and minority stake acquisitions for all available countries during our sample period. Failed deals are defined as those deals announced in a given year that are not completed as of the end of our sample period. Such deals are either withdrawn, have "unknown status," or are still pending years after their initial announcement (suggesting perhaps regulatory barriers to completion). Based on these data, we follow the approach taken in constructing our *Cross-border ratio* to construct *Failed bank deals* as the sum of all failed bank M&A deals in year t involving targets from country i and all failed bank deals involving acquirers from country j , as a proportion of all deals in target country i in year t . We construct *All failed deals* in a similar manner, but use the total number of failed deals (including those involving nonfinancial firms).³⁹

To the extent that these variables capture changes in the tightness of merger controls, they should have a negative impact on cross-border bank flows. Before including these variables as controls in our regressions, we examine whether they are important factors in determining cross-border bank acquisition activity. To do so, we run regressions of our *Cross-border ratio* on a set of controls excluding our main regulatory variables, and then examine the incremental explanatory role of each of these variables separately. The coefficients on these two proxies are negative and statistically significant, suggesting that they capture at least some of the tightness of merger controls. We report results from OLS and PPML regressions of cross-border bank acquisition flows on these and other proxies for merger control (discussed later) in the Internet Appendix.

³⁹ One concern with our measure of merger control is that failed deals may have increased significantly toward the end of our sample period. We examine whether this is in fact the case. The total number of failed bank deals declined throughout the first half of our sample period, and, while the number of failed deals increased in the latter half of our sample period, the levels remain well below those at the start of the sample period. The Internet Appendix provides graphs of the number of failed bank deals by year and the average ratio of failed bank deals by year.

We use *Failed bank deals* as our main proxy for the tightness of merger control, as it should capture any bank deals that are announced and fail to be completed (plausibly, because of intervention by home or host country regulators). Admittedly, this variable does not capture all plausible bank deals that could have occurred without intervention by regulatory authorities. Nonetheless, it should capture some of these changes in the tightness of merger controls.

We present results from PPML estimation of equation (4) including our main proxy for merger control in Table VII. To conserve space, we do not report the coefficients on the remaining control variables.⁴⁰ The results confirm that, by itself, our proxy for merger control is negatively associated with cross-border flows as expected. However, looking at the next five specifications in the table, we note that the regulatory variables do not lose their significance even after accounting for changes in merger control using this proxy. The magnitude of the coefficients on the differences in regulatory variables and their economic significance are comparable to those reported in Panel A of Table IV. For example, a one- σ increase in the difference in activities restrictions (2.96 units) is associated with a 1.18% increase in the *Cross-border ratio*, or 43% of its σ .⁴¹ Using the alternate measure of merger control, which should capture changes in the tightness of merger controls in a country that are not specific to the banking sector, yields similar results (see the Internet Appendix). All regulatory variables retain their positive and significant coefficients when using this alternate proxy for merger control.

We also test the robustness of our results using alternate proxies for merger control and report these results in the Internet Appendix. First, we use the fraction of foreign bank applications that are denied, based on the surveys from Barth, Caprio, and Levine (2013). The results corroborate our findings; the magnitudes of the coefficients on the regulatory variables are actually larger in several specifications. Second, we use the Competition Policy Index from Buccirossi et al. (2011) as another proxy. While this variable is available annually, it is only available for 12 target countries and thus, when we use this proxy, our sample size is vastly reduced and we lose power and precision in our estimates. None of the coefficients of interest are significant in these regressions, but the regulatory variables retain their positive sign.

As another way to examine whether merger control policy affects our results, we study deals that are less likely to receive regulatory scrutiny. In particular, we rerun our analysis using an alternate *Cross-border ratio* that includes cross-border acquisitions also involving minority stakes. Consistent with the view that merger control policies are less of a factor in smaller deals involving minority acquisitions, our results are stronger when we include such deals.

⁴⁰ Full results from these regressions (including coefficients on the controls) are presented in the Internet Appendix.

⁴¹ Recalling that the coefficients in PPML should be interpreted as if the dependent variable is in logs, given the coefficient on *Restrictions on bank activities* in Table VII (0.579), a one-SD increase (2.96 units) is associated with a 5.55 times ($e^{0.579 \times 2.96}$) increase in the mean *Cross-border ratio* from 0.26% to 1.44%, or an increase of 43% of its σ (2.78).

Table VII
Assessing the Impact of Merger Control on Cross-Border Bank Acquisitions

The table shows estimates from panel regressions of majority cross-border bank acquisitions by country-pair-year using the PPML estimator following Santos Silva and Tenreyro (2006). The dependent variable is the annual *Cross-border ratio* for each country pair—the total number of majority cross-border bank acquisitions in year t in which the target is from country i and the acquirer is from country j ($i \neq j$), as a proportion of all domestic and cross-border bank acquisitions in target country i in year t . If a target country does not have any cross-border bank acquisition activity in a given year, we exclude that country for that year. Independent variables are computed as differences between acquirer country j and target country i . As a proxy for merger control, we include a measure of failed bank deals—the sum of all failed bank M&A deals involving targets in country i in year t , plus all failed bank deals involving acquirers in country j in year t , as a proportion of all deals in target country i in year t . See Appendix B for a description of the four indices of regulatory quality from the revised database from Barth, Caprio, and Levine (2013) and for a description of the independent variables. See notes to Table III on Hansen's J -statistic. To conserve space, we do not report the coefficients on the control variables. We include acquirer country, target country, and year fixed effects (not shown) and cluster standard errors by target country. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: Cross-Border Ratio _{<i>i,j</i>}						
	No IVs						No U.S. targets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ restrictions on bank activities $j-i$		0.579** (2.42)					
Δ stringency of capital regulation $j-i$			1.393** (2.56)				
Δ official supervisory power $j-i$				0.400* (1.79)			
Δ private monitoring $j-i$					4.489*** (3.77)		
Δ regulation overall $j-i$						4.826* (1.72)	16.051* (1.67)
Failed bank deals	−0.426** (−2.56)	−0.489*** (−2.82)	−0.481** (−2.50)	−0.369** (−2.11)	−0.283 (−1.56)	−0.524** (−2.10)	−0.186 (−0.70)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Target country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,332	23,732	22,842	24,332	23,253	22,651	20,464
Pseudo R^2	0.057	0.054	0.017	0.045	0.017	0.008	0.093
Hansen J -statistic (χ^2)		1.49	7.63	21.38	3.03	0.079	12.67
p -value		0.47	0.02	0.00	0.22	0.78	0.00

We report the results including minority acquisitions in the Internet Appendix with and without our main proxy for merger control.

Overall, the results in this section show that merger control policies do influence the pattern of cross-border bank acquisition flows. More importantly, the results confirm our prior findings that the flow of cross-border bank acquisitions is positively related to differences in regulatory quality between acquirer and target countries. These results continue to hold even after accounting for the impact of merger control in various ways. We acknowledge, however, that we cannot perfectly control for this important determinant of cross-border bank acquisition flows.

B. Impact of Merger Control Policies on Share Price Reactions to Cross-Border Bank Acquisitions

Next, we examine how the tightness of merger control policies affects share price reactions to cross-border bank acquisition announcements. To do so, we re-estimate equation (7) including our main proxy for merger control, *Failed bank deals*. We report results from this analysis in Table VIII. To conserve space, we omit the coefficients on the control variables.

The results continue to confirm our prior findings. Target bank CARs are higher when the acquirer comes from a country with more restrictions on bank activities, stricter capital requirements, stronger private monitoring, and better regulatory quality. While differences in *Official supervisory power* are positively correlated with target CARs, the result is not statistically significant. Acquirer bank CARs are also higher for deals involving acquirers from countries with better regulatory quality—see Model (6). Differences in overall regulatory quality are also positively correlated with aggregate CARs, although the result is not statistically significant. The magnitudes of the results in Table VIII are similar to those reported in Table VI. From Model (1), we observe that a one- σ increase in the *stringency of capital regulation* index is associated with a 0.67% higher target CAR, a substantial difference compared to the 5.8% CAR overall. Overall, our results hold even after controlling for differences in merger control policy.

The results in Table VIII also show that target (and acquirer) CARs are positively related to the merger control proxy. Deals involving countries with tighter merger controls (as indicated by a higher proportion of failed bank deals as of the prior year-end) are associated with larger CARs, all else being equal. These results are both statistically and economically significant. A one- σ increase in the proportion of failed bank deals (0.28) is associated with 0.15% higher CARs. One may interpret this more favorable reaction to cross-border deals in the face of tough barriers as consistent with the finding in Carletti, Hartmann, and Ongena (2013) that shows bank stock prices react positively to the introduction of legislative reforms in merger control legislation in 19 industrial countries between 1987 and 2004. They interpret their results as evidence that merger control policies include “checks and balances” that

Table VIII
Regressions of Cumulative Abnormal Returns with Merger Controls

The table shows results from 2SLS regressions of CARs for targets and acquirers in domestic and cross-border bank acquisitions over the period 1995 to 2012. The dependent variable represents CARs for targets (targets and acquirers) over the period $t = (-2, +2)$ around the announcement day. See notes to Table V for a description of the estimation of CARs. All returns for international banks are obtained from DataStream, while returns for U.S. institutions are obtained from CRSP. Independent variables are computed as differences between acquirer country j and target country i . As a proxy for merger control, we include a measure of failed bank deals—the sum of all failed bank M&A deals involving targets in country i in year t and all failed bank deals involving acquirers in country j in year t , as a proportion of all deals in target country i in year t . See Appendix B for a description of the indices of regulatory quality from the revised database from Barth, Caprio, and Levine (2013) as well as the additional control variables. To address endogeneity concerns, we instrument the regulatory variables using the government ownership of banks in 1970, the number of systemic banking crises in the 1970s and 1980s, and the percentage number of years since 1776 that each country has been independent. To conserve space, we do not show the coefficients on the additional control variables. We include acquirer country, target country, and year fixed effects (not shown) in all regressions and cluster standard errors by target country. Additional details on regression diagnostics are furnished in the caption for Table III. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: Target CARs (-2, +2)					Acquirer CARs	Aggregate CARs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ restrictions on bank activities $j-i$	0.123*** (7.77)						
Δ stringency of capital regulation $j-i$		0.561*** (3.88)					
Δ official supervisory power $j-i$			0.056 (1.11)				
Δ private monitoring $j-i$				0.037*** (3.66)			
Δ regulation overall $j-i$					0.070** (2.57)	0.078** (2.41)	0.037 (1.63)

(Continued)

Table VIII—Continued

	Dependent Variable: Target CARs (-2, +2)				Acquirer CARs	Aggregate CARs	
	(1)	(2)	(3)	(4)			(5)
Failed bank deals	0.549*** (3.59)	0.513*** (3.38)	0.535*** (3.52)	0.555*** (3.62)	0.556*** (3.60)	0.117*** (3.63)	0.117*** (2.30)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Target country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	867	866	867	867	866	1,256	623
Adjusted R^2	0.222	0.215	0.218	0.224	0.224	0.119	0.220
Partial R^2	0.90	0.05	0.18	0.60	0.21	0.05	0.69
First-stage F -statistic	102.06	22.67	59.35	79.28	24.09	3.57	657.91
p -value	0.00	0.00	0.00	0.00	0.00	0.04	0.00
Sargan χ^2 statistic	4.56	0.68	4.74	2.54	0.07	0.14	1.84
p -value	0.10	0.41	0.09	0.11	0.79	0.70	0.40

mitigate potential abuse and wasteful enforcement of supervisory control in the banking sector.

C. Push and Pull Factors in Cross-Border Bank Acquisitions

To further assess the motives behind regulatory arbitrage in cross-border deals, we next examine whether such flows are driven by push or pull factors. Our evidence on cross-border bank flows shows increased cross-border acquisitions flows from acquirers with stronger regulatory regimes, in line with the findings on international bank flows in Houston, Lin, and Ma (2012). But Houston, Lin, and Ma (2012) uncover additional evidence that bank flows are influenced as much by the tougher regulatory restrictions in the acquirer's home country as by the weaker regulatory regime in the target's home country. That is, both push and pull factors are at work. They interpret these findings as evidence of a detrimental race to the bottom in bank regulation.

If regulatory arbitrage in cross-border bank acquisition flows is driven by this harmful motive, we should also find that the flows are driven by both push and pull factors. If, on the other hand, acquirer banks from tough regimes are engaging in cross-border bank acquisitions to exploit opportunities elsewhere due to excessively burdensome regulation at home, then our results should be primarily driven by push factors. In this case, acquirers may not necessarily target banks in weaker regimes, which would be inconsistent with the harmful race-to-the-bottom view of regulatory arbitrage.

To test the impact of push and pull factors, we re-estimate equation (4) using levels of regulatory quality in the acquirer and target countries, instead of differences. As before, we instrument these regulatory variables to address endogeneity concerns. We present results from PPML estimations in Table IX. To conserve space, we only report results using the overall regulation index, *Regulation overall*. The results in Table IX show that push factors influence cross-border flows to a much greater extent than pull factors, which are not significant. The results are both statistically and economically significant. In Model (1), a one-SD increase in the *Regulation overall* index (0.16) in the acquirer country is associated with a 0.65% increase in the *Cross-border ratio*, or 23.2% of its σ .⁴² In contrast, the target's overall regulatory index is not significantly associated with cross-border bank acquisition flows. The coefficient on the target's overall regulatory index is actually positive, albeit statistically insignificant, which does not square with the destructive form of regulatory arbitrage. When we include the overall regulatory quality for both the target and the acquirer's countries in the regressions (Model (3)), our results continue to show that push factors drive cross-border bank acquisitions.⁴³

⁴² Coefficients in PPML should be interpreted as if the dependent variable is in logs. Thus, given the coefficient on *Regulation overall* in Model (1) of Table IX (7.80), a one- σ increase (0.16 units) is associated with a 3.48 times ($e^{7.8 \times 0.16}$) increase in the *Cross-border ratio* from 0.26% to 0.91%, or an increase of 23.2% of its σ (2.78).

⁴³ We present results for the full set of regulatory variables in the Internet Appendix. When using the acquirer's regulatory variables, four of the five (*Restrictions on bank activities* is the

Table IX
Cross-Border Bank Acquisition Flows: Push and Pull Factors

The table shows estimates from panel regressions of majority cross-border bank acquisitions by country-pair-year using the PPML estimator following Santos Silva and Tenreyro (2006). The dependent variable is the annual *Cross-border ratio* for each country pair—the total number of majority cross-border bank acquisitions in year t in which the target is from country i and the acquirer is from country j ($i \neq j$), as a proportion of all domestic and cross-border bank acquisitions in target country i in year t . If a target country does not have any cross-border bank acquisition activity in a given year, we exclude that country for that year. Independent variables are computed as differences between acquirer country j and target country i . See Appendix B for a description of the index of regulatory quality derived from the revised database from Barth, Caprio, and Levine (2013) and for a description of the independent variables. In Models (4) and (5), we interact the acquirer's regulatory quality with indicators of the target country's institutional quality using: (1) the index of creditor rights from Djankov, McLiesh, and Shleifer (2007) and (2) the rule of law index from Kaufmann, Kraay, and Mastruzzi (2009). We first average each institutional quality variable for each target country over time. The institutional quality indicators equal one for target countries with average institutional quality above the median and zero otherwise. Definitions of all variables are found in Appendix B. See notes to Table III on Hansen's J -statistic. We include acquirer country, target country, and year fixed effects (not shown) and cluster standard errors by target country. Heteroskedasticity-robust t -statistics are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: Cross-Border Ratio $_{i,j}$				
	(1)	(2)	(3)	(4)	(5)
Regulation overall $_j$	7.800** (2.37)		2.695** (2.33)	2.745** (2.35)	6.065* (1.83)
Regulation overall $_i$		0.686 (1.06)	0.159 (0.39)	0.159 (0.40)	0.085 (0.19)
Regulation overall $_j$ × creditor rights $_i$				−0.536 (−1.30)	
Regulation overall $_j$ × rule of law $_i$					−4.205 (−1.59)
Δ governance index $_{j-i}$	2.144*** (3.52)	0.718* (1.77)	0.583 (0.75)	0.618 (0.80)	0.581 (0.63)
Δ log GDP per capita $_{j-i}$	0.761 (1.27)	0.828** (2.29)	0.719 (1.19)	0.723 (1.21)	0.334 (0.48)
Δ GDP growth $_{j-i}$	−1.927 (−0.64)	2.196 (1.03)	−0.175 (−0.06)	−0.147 (−0.05)	0.007 (0.00)
Δ bank credit $_{j-i}$	−0.228 (−1.13)	−0.004 (−0.10)	−0.188 (−1.56)	−0.187 (−1.55)	−0.160 (−1.20)
Δ bank concentration $_{j-i}$	0.011 (1.40)	−0.001 (−0.18)	0.014** (2.06)	0.014** (2.07)	0.014* (1.95)
Δ real exchange rate return $_{j-i}$	0.065 (0.17)	−0.345 (−1.25)	−0.101 (−0.26)	−0.096 (−0.24)	0.021 (0.04)
Δ real stock market return $_{j-i}$	0.248 (0.70)	0.048 (0.37)	0.254 (0.82)	0.261 (0.84)	0.434 (1.19)
Same language	1.291*** (4.80)	0.989*** (5.93)	1.093*** (4.27)	1.176*** (4.38)	1.847*** (2.71)
Colony	0.623* (1.89)	1.178*** (6.15)	0.702** (2.11)	0.776** (2.23)	1.942** (2.06)
Contiguous	0.096 (0.27)	−0.204 (−0.94)	0.163 (0.46)	0.132 (0.36)	−0.434 (−0.79)

(Continued)

Table IX—Continued

	Dependent Variable: Cross-Border Ratio _{ij}				
	(1)	(2)	(3)	(4)	(5)
Distance	−0.966*** (−6.65)	−0.939*** (−11.53)	−1.009*** (−6.67)	−0.995*** (−6.56)	−1.200*** (−4.29)
Bilateral trade	−0.458 (−0.41)	0.441 (0.68)	−1.109 (−1.06)	−1.019 (−0.96)	−0.089 (−0.06)
Δ growth opportunities $j-i$	1.926 (0.71)	−1.242 (−1.13)	0.475 (0.20)	0.412 (0.18)	0.433 (0.16)
Target country f.e.	Yes	Yes	Yes	Yes	Yes
Acquirer country f.e.	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes
Observations	22,651	22,651	22,651	22,651	22,651
Pseudo R^2	0.019	0.083	0.030	0.029	0.016
Hansen's J -statistic (χ^2)	3.72	12.54	0.00	0.00	0.55
p -value	0.16	0.00	0.99	0.98	0.46

As a final test of the destructive race-to-the-bottom form of regulatory arbitrage, we examine whether acquirers from tough regulatory regimes specifically target countries with lower institutional quality. To do so, we use two measures of target country institutional quality: the creditor rights index from Djankov, McLiesh, and Shleifer (2007) and the rule of law index from Kaufmann, Kraay, and Mastruzzi (2009). Using these variables, we create indicators of institutional quality that equal one if the target country's institutional quality is above the cross-country median, and zero otherwise. Models (4) and (5) in Table IX present the results. We focus on the coefficients on the interactions between the acquirer's overall regulation index and these two indicators of institutional quality. The interaction terms are both statistically insignificant, suggesting that target institutional quality is not a driving force behind cross-border bank acquisitions.

Our results in this section provide additional evidence more consistent with a benign motive for regulatory arbitrage in cross-border bank acquisitions than with a destructive, race-to-the-bottom motive. In particular, the findings suggest that acquirers from stronger regulatory regimes are not necessarily targeting countries with weaker regimes when making cross-border bank acquisitions.

V. Additional Robustness Tests

We perform additional tests to verify the robustness of our results and present the results in the Internet Appendix. First, with respect to the cross-sectional regressions in Table III, we show results using alternate estimation methods, namely, 2SLS and SMM. In addition, we examine the robustness of

exception) variables have positive and statistically significant coefficients, while none of the target's regulatory variables are significant.

our findings to computing the *Cross-border ratio* as the current dollar *value* of majority cross-border bank acquisitions between country pairs scaled by the total *value* of domestic and cross-border acquisitions in the target country over the period. We also present results from PPML estimations using the raw regulatory variables, and for PPML and SMM tests using the raw count of cross-border deals (retaining instrumental variables for the regulatory indexes as in Table III). Finally, we evaluate a number of alternative indexes from Barth, Caprio, and Levine (2013) and find similarly robust evidence of acquisition flows consistent with regulatory arbitrage. Overall, our main findings continue to hold when using these alternate estimation methods and specifications: cross-border acquisition flows tend to involve acquirers from stronger regulatory regimes.⁴⁴

Next, we perform a number of robustness tests for the panel regression analysis of Table IV. In particular, we estimate OLS regressions with target-country-year and acquirer-country-year fixed effects using the raw regulatory variables and we estimate PPML regressions with raw regulatory indexes. Our key findings continue to hold after including additional control variables from the trade literature, as well as controls for the number of greenfield banks from the acquiring country that operate in the target country as of the beginning of the year, using data on foreign bank ownership from Claessens and van Horen (2014). In addition, though the United States represents an overwhelmingly popular target market for cross-border deals, our findings on flows in the direction of regulatory arbitrage remain when we exclude U.S. banks as targets or acquirers. Finally, we test the robustness of our results to the inclusion of controls for the quality of information sharing in the target country, which has been shown to affect cross-border bank entry (Bouckaert and Degryse (2006), Giannetti, Jentzsch, and Spagnolo (2010)). Our results continue to go through.

An alternative explanation for our results is that regulators in tightly regulated environments are reluctant to allow acquisitions by banks from weakly regulated countries. If this is the case, we should observe, as we do, that acquirers are predominantly from tougher environments (perhaps because acquisitions in the other direction are blocked). To test this conjecture, we construct the number of failed bank deals in which the acquirer is from country i and the target is from country j ($i \neq j$), as a proportion of all deals in the target country i in year t . We use this cross-border failed bank deals ratio as a dependent variable in the panel regressions. If regulators from tougher regimes are more likely to block deals in which acquirers are from countries with weaker regulatory quality, we expect to observe negative and significant coefficients on the differences in the regulatory variables. We do not find any evidence of this in our sample. All the regulatory variables have positive coefficients, and one—*Private monitoring*—is significant at the 10% level, suggesting that there are actually more failed cross-border bank deals when acquirers come from countries with stronger private monitoring.

⁴⁴ We report these results in Panel C of Table 4A of the Internet Appendix.

Finally, we use the bid premiums as a dependent variable in our cross-sectional regression tests for the subset of deals with available bid premium data. In four of the five specifications, we find a reliably positive coefficient on the differences in regulatory indexes (*Restrictions on bank activities* has the wrong sign). We believe this finding is consistent with the main results on target CARs and is not surprising given the findings in Schwert (2000) linking the two as proxies for merger premiums.

VI. Conclusion

This paper explores an important yet understudied aspect of cross-border bank acquisitions, namely, the impact of differences in national bank regulations on the level of deal activity and on the shareholder wealth created around deal announcements. In particular, we focus on whether regulatory arbitrage is a driving force behind the increased cross-border bank acquisition activity observed over the last decade or so. We test two competing hypotheses on regulatory arbitrage. On the one hand, regulatory arbitrage could be driven by the search for profitable opportunities if banks from overly restrictive regimes are not allowed to engage in certain risky but value-generating activities. Through bank acquisitions of targets in countries with fewer regulatory restrictions, such banks may escape from costly regulations and transfer sound supervision from the home country. On the other hand, a more harmful view of regulatory arbitrage holds that it could lead to a race-to-the-bottom if banks target countries with weak regulations in order to pursue risky, value-destroying activities. We find evidence more in favor of the former than the latter hypothesis.

Using a sample of 916 (7,297) majority cross-border (domestic) acquisitions announced between 1995 and 2012, we find that differences in bank regulation do affect cross-border bank acquisition flows and share price reactions to acquisition announcements. In particular, acquirers are typically from countries with stronger supervision, more restrictions on bank activities, stricter capital requirements, and stronger private monitoring, all of which is in line with regulatory arbitrage. The quality of bank regulation plays an economically important role in explaining cross-border acquisition flows over and above a number of alternative country-level factors, including time-varying differences in merger control policies, despite the possibility of some forms of endogeneity and reverse causality. This result is robust to different estimation approaches. Furthermore, in line with the more benign view of regulatory arbitrage, we find that our results are driven mostly by push factors, as acquirers come from tougher regulatory regimes but do not necessarily target countries with weaker regimes.

We find that target banks' CARs and aggregate CARs around the announcement date are positively correlated with differences in the quality of bank regulation between the acquirer and target countries. Target (and aggregate) abnormal returns are higher when acquirers are from countries with a tougher bank regulatory environment. In particular, target (aggregate) CARs are higher when acquirers are from countries with more restrictions on bank activities,

stricter capital requirements, stronger private monitoring, and better overall regulatory quality.

Caveats abound in our study, but two are worthy of note. A distinct advantage of our effort relative to others is that we evaluate regulatory arbitrage *at the deal level*, studying the market's reaction to deal announcements to assess the economic consequences of regulatory arbitrage and measuring the attributes of acquirers and targets to understand the motives behind it. But with this advantage comes a cost. Our effort narrowly frames the issue of regulatory arbitrage in the context of cross-border bank acquisitions, which is but one of many important types of multinational banking activities that may be affected by regulation. Furthermore, our sample period is short (starting in 1995). To some extent we are constrained by data availability, but more importantly cross-border bank acquisitions were fewer in number, smaller in size, and limited in scope by many government rules that blocked foreign acquisitions until recently.

Our research is important given the renewed focus on regulation in the global banking sector in the aftermath of the recent global financial crisis. Indeed, the increased importance of regulations and governance mechanisms in the banking industry is highlighted in the Basel Committee's December 2009 Report on Strengthening the Resilience of the Banking Sector, which stresses the vital role governance mechanisms, transparency, and disclosure can play in promoting stability in the banking sector. The impact of these rules and other proposed regulatory changes that would lead to more stringent government oversight of financial institutions throughout the world will certainly have an impact on banks and in turn on the financial sector as a whole. A major driver behind this push for tougher regulations is concern about regulatory arbitrage. Our findings show that not all forms of regulatory arbitrage may be a cause for concern, at least within the scope of cross-border acquisitions.

Initial submission: January 24, 2012; Final version received: January 25, 2015
Editor: Kenneth Singleton

Appendix A
Regulatory Variables by Country

Listed below are measures of bank regulation. We use four indices of the quality of regulation from the updated database from Barth, Caprio, and Levine (2013), built from data gathered from surveys conducted by the World Bank. These include: (1) *Restrictions on bank activities*, which measures regulatory impediments to banks engaging in securities market activities (underwriting, brokering, dealing, mutual funds), insurance activities (underwriting, selling), and real estate (development, management); (2) *Stringency of capital regulation*, which captures how much capital banks must hold, as well as the sources of funds that count as regulatory capital; (3) *Official supervisory power*, which measures whether supervisory authorities have the power to take actions to prevent or correct problems; and (4) *Private monitoring*, which measures whether there are incentives for the private monitoring of banks. Finally, *Regulation overall* is the first principal component of the four regulatory indices. Values for all indices are averaged across all four surveys for each country. Higher values of the indices indicate more restrictions on bank activities, more stringent capital requirements, stronger supervisory power, higher private monitoring, and better regulatory quality, respectively.

Country	Restrictions on Bank Activities		Stringency of Capital Regulation	Official Supervisory Power	Private Monitoring	Regulation Overall	Country	Restrictions on Bank Activities	Stringency of Capital Regulation	Official Supervisory Power	Private Monitoring	Regulation Overall
Argentina	7.11	7.11	9.83	8.72	0.13	Mauritius	9.83	7.00	10.78	8.67	0.34	
Australia	7.22	6.94	11.56	10.00	0.99	Mexico	7.44	6.78	11.28	7.89	-0.55	
Austria	4.50	7.33	12.06	6.22	-0.64	Morocco	8.83	6.06	12.11	8.00	0.67	
Bahrain	7.78	5.33	13.36	9.28	1.05	Namibia	6.89	4.58	11.00	8.39	0.17	
Bangladesh	9.78	3.94	11.47	7.17	0.07	Netherlands	4.61	6.44	8.36	8.44	-1.02	
Belgium	6.22	6.89	11.17	7.11	-0.34	New Zealand	3.50	2.83	9.12	9.89	-0.51	
Brazil	6.67	5.39	13.67	8.89	0.88	Nigeria	7.61	6.61	12.11	8.22	0.55	
Bulgaria	7.17	6.94	11.39	7.50	0.03	Norway	6.36	7.55	9.46	7.76	-0.88	
Canada	5.00	4.22	7.67	8.78	-0.73	Oman	9.00	6.22	13.06	7.94	0.94	
Chile	9.00	5.61	10.94	7.44	0.14	Pakistan	9.27	8.18	13.89	8.82	1.08	
China	10.78	5.00	10.79	8.94	1.11	Peru	6.89	4.50	12.22	7.72	0.57	
Colombia	9.64	5.50	13.36	8.82	1.18	Philippines	5.00	5.50	11.81	8.67	0.36	
Croatia	6.00	4.94	11.56	7.55	-0.33	Poland	7.61	5.78	9.91	7.89	-0.17	
Cyprus	6.72	5.92	11.78	7.72	0.08	Portugal	6.94	6.44	13.36	7.06	0.34	
Czech Republic	7.44	4.44	10.72	6.67	-0.17	Qatar	5.50	6.11	12.56	9.89	1.63	
Denmark	6.22	5.94	9.11	9.17	-0.32	Romania	8.56	5.00	10.07	6.11	-0.72	
Ecuador	9.11	9.00	14.06	10.00	1.87	Russia	5.89	7.06	8.89	6.78	-0.95	
Egypt	8.28	5.56	13.11	8.89	1.00	Saudi Arabia	8.39	4.39	13.61	10.78	1.63	
Estonia	5.33	6.06	12.33	8.00	0.59	Singapore	6.39	7.33	12.09	9.22	0.69	

(Continued)

Appendix A—Continued

Country	Restrictions on Bank Activities	Stringency of Capital Regulation	Official Supervisory Power	Private Monitoring	Regulation Overall	Country	Restrictions on Bank Activities	Stringency of Capital Regulation	Official Supervisory Power	Private Monitoring	Regulation Overall
Finland	5.78	5.22	7.50	8.89	-0.55	Slovak Republic	7.44	5.89	12.20	6.44	-0.09
France	5.28	6.11	7.86	7.00	-1.32	Slovenia	7.00	7.56	13.56	7.89	0.76
Germany	4.22	6.50	9.11	7.39	-1.24	South Africa	6.94	7.11	6.89	9.61	-0.53
Ghana	9.06	7.39	11.28	6.50	0.16	South Korea	7.28	5.33	10.00	10.39	0.29
Greece	6.83	5.39	10.22	7.28	-0.73	Spain	5.39	8.89	9.97	8.39	-0.34
Hong Kong	3.22	5.45	10.95	8.73	-0.57	Sri Lanka	6.72	6.44	9.88	8.50	-0.04
Hungary	7.17	6.83	14.03	8.11	0.81	Sweden	7.00	3.00	7.22	7.00	-1.32
Iceland	7.44	5.39	7.11	7.61	-1.15	Switzerland	4.28	6.28	13.28	7.72	0.34
India	8.72	6.83	9.22	7.22	-0.61	Taiwan	9.61	4.44	11.56	8.00	0.23
Indonesia	10.22	6.39	14.11	8.50	1.41	Thailand	8.00	5.11	10.38	8.44	-0.01
Ireland	5.28	5.47	9.94	9.72	0.04	Trinidad	7.67	4.61	8.28	7.17	-0.76
Israel	9.72	6.06	9.02	9.50	0.29	Tunisia	8.18	8.36	12.94	6.64	-0.05
Italy	7.78	4.56	7.28	7.22	-1.22	Turkey	7.50	5.67	13.36	7.61	0.63
Japan	8.78	5.39	11.97	9.00	0.72	Ukraine	5.00	6.55	12.07	8.00	0.29
Jordan	7.78	7.83	10.47	7.28	-0.24	UAE	5.73	7.18	13.44	11.00	2.44
Kazakhstan	6.89	5.44	11.55	6.78	-0.28	United Kingdom	3.72	6.94	9.83	10.00	-0.37
Kenya	8.67	6.06	13.61	7.33	0.95	United States	8.50	6.28	13.42	9.33	1.20
Kuwait	6.50	7.11	10.50	10.61	0.78	Venezuela	8.61	3.83	12.67	5.72	0.47
Lithuania	7.22	4.28	11.19	8.06	0.07	Vietnam	12.00	7.00	11.45	.	.
Luxembourg	4.94	7.00	12.17	7.72	0.25	Average	7.18	5.99	11.16	8.21	0.16
Malaysia	7.39	3.67	13.55	9.00	1.03	SD	1.74	1.25	1.93	1.16	0.80

Appendix B

Variable Definitions

Variable	Definition
<i>Cross-border ratio_{i,j}</i>	The total number of majority cross-border bank acquisitions in year t in which the target is from country i and the acquirer is from country j ($i \neq j$), as a proportion of all domestic and cross-border bank acquisitions in target country i in year t . The ratio is calculated annually for each country pair. We exclude a country i in years in which it did not have any cross-border bank acquisition activity. Source: Thomson Reuters's SDC.
<i>Restrictions on bank activities</i>	Index measuring regulatory impediments to banks engaging in securities market activities, insurance activities, and real estate activities. The index ranges from 3 to 12, with higher values indicating more restrictiveness. Data obtained from Barth, Caprio, and Levine (2013).
<i>Stringency of capital regulation</i>	Index measuring the stringency of regulations regarding how much capital banks must hold, as well as the sources of funds that count as regulatory capital. The index ranges from 0 to 10, with higher values indicating greater stringency. Data obtained from Barth, Caprio, and Levine (2013).
<i>Official supervisory power</i>	Index measuring whether supervisory entities have authority to take action to prevent and correct problems. The index ranges from 0 to 14, with higher values indicating greater power. Data obtained from Barth, Caprio, and Levine (2013).
<i>Private monitoring</i>	Index measuring whether there exist incentives/ability for the private monitoring of banks. The index ranges from 0 to 12, with higher values indicating more private oversight. Data obtained from Barth, Caprio, and Levine (2013).
<i>Regulation overall</i>	Index of overall regulatory quality that is the first principal component of the four Barth, Caprio, and Levine (2013) indices: <i>Restrictions on bank activities</i> , <i>Stringency of capital regulation</i> , <i>Official supervisory power</i> , and <i>Private monitoring</i> . We construct the index from each of the four regulatory surveys. A detailed description of the principal component analysis is provided in the Internet Appendix.
<i>Governance index</i>	The average of all six Kaufmann, Kraay, and Mastruzzi (2009) governance indicators: political stability, voice and accountability, government effectiveness, regulatory quality, control of corruption, and rule of law. Each of the indices ranges from -2.5 to 2.5 , with higher values indicating better governance.
<i>GDP growth</i>	Annual growth in real GDP. Source: World Development Indicators.
<i>Log GDP per capita</i>	Logarithm of real GDP (current US\$) divided by the average population. Source: World Development Indicators.

(Continued)

Variable Definitions—Continued

Variable	Definition
<i>Bank concentration</i>	Assets of the three largest banks as a share of all commercial banks' assets. Source: Global Financial Development Database (Beck, Demirgüç-Kunt, and Levine (2009)).
<i>Bank credit (log)</i>	Logarithm of total credit provided by deposit money banks to the private nonfinancial sector (in \$billion). Source: Bank for International Settlements (BIS). For countries for which data are not available from BIS, the total bank credit to the private sector (in \$billion) is derived from the ratio of bank credit to GDP from the World Bank's Global Financial Development Database (Beck, Demirgüç-Kunt, and Levine (2009)) updated in Čihák et al. (2012). Values are converted into US\$ using end-of-year exchange rates (WMR/Reuters).
<i>Real exchange rate return</i>	Annual (prior 12-month) real bilateral U.S. dollar exchange rate return. Exchange rates (WMR/Reuters) and 2000 constant dollar Consumer Price Index data are obtained from Thomson Reuters's DataStream.
<i>Real stock market return</i>	Annual (prior 12-month) real stock market return. The DataStream local currency country-level stock market return indices are deflated using the Consumer Price Index (base 2000).
<i>Growth opportunities</i>	A measure of a country's exogenous growth opportunities following Bekaert et al. (2007). It is the log of the inner product of the vector of global industry price-to-earnings ratios and the vector of country-specific industry weights. Source: DataStream.
<i>Bilateral trade</i>	Maximum of bilateral imports and exports between two countries. Bilateral imports (exports) are calculated as the total value of imports (exports) by a target's country from an acquirer's country as a proportion of total imports by the target's country. Source: IMF's Direction of Trade Statistics.
<i>Failed bank deals</i>	Failed bank deals as a proportion of all deals in the target country. This variable equals the sum of all failed bank M&A deals involving targets in country i in year t , plus all failed bank deals involving acquirers in country j in year t , as a proportion of all deals in target country i in year t . Failed bank deals are those bank M&A deals announced in year t that are not completed as of the end of our sample period. These include deals that are withdrawn, deals that are still pending years after the announcement, as well as deals with unknown status as of the end of our sample period. Source: Thomson's SDC Platinum.
<i>NPL-to-GL</i>	Total nonperforming loans (past due 90 days or more) as a share of gross loans. Source: WorldScope and Bloomberg.
<i>ROA</i>	Net income plus interest expense divided by total assets. Source: WorldScope and Bloomberg.
<i>Z-score</i>	$(ROA + \text{equity/assets})/\sigma(ROA)$, where the σ of ROA, $\sigma(ROA)$, is estimated as a three-year moving average. Source: WorldScope and Bloomberg.
<i>Size</i>	Total assets (log). Source: WorldScope and Bloomberg.

(Continued)

Variable Definitions—Continued

Variable	Definition
<i>Contiguous</i>	Indicator variable equal to one if the two countries share a border and zero otherwise. Source: Mayer and Zignago (2011).
<i>Same language</i>	Indicator variable equal to one if the two countries share the same language and zero otherwise. Source: Mayer and Zignago (2011).
<i>Distance</i>	Log of the circle distance (in km) between the countries' capitals. Source: Mayer and Zignago (2011).
<i>Colony</i>	Indicator variable equal to one if the two countries have ever had a colonial link and zero otherwise. Source: Mayer and Zignago (2011).
<i>Rule of law index</i>	Index that captures perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. Source: Kaufmann, Kraay, and Mastruzzi (2009).
<i>Creditor rights</i>	Index aggregating creditor rights. Specifically, it indicates: (1) whether restrictions, such as creditors' consent or minimum dividend, are imposed when a creditor files for reorganization, (2) whether there is no automatic stay or asset freeze imposed by the court to limit a creditor's ability to seize collateral, (3) whether secured creditors are ranked first in the distribution of proceeds in the liquidation of a bankrupt firm, and (4) whether existing management does not remain in control of the firm during the reorganization. Source: Djankov, McLiesh, and Shleifer (2007).
<i>CARs</i>	Cumulative abnormal returns around the announcement of the acquisitions. We obtain abnormal returns by estimating a market model using a world market index from 260 to 21 days prior to the announcement of the acquisition.

Appendix C

Summary Statistics on Main Variables

The table shows summary statistics of the main variables in Panel A and correlations between the variables in Panel B. *Cross-border ratio*_{*i,j*} is the annual measure of the number of majority cross-border bank acquisitions in year *t* where the target is from country *i* and the acquirer is from country *j*, as a proportion of all bank acquisitions in target country *i* in year *t*. *Restrictions on bank activities* is an index that measures regulatory impediments to banks engaging in securities, insurance, and real estate activities, *Stringency of capital regulation* is an index that measures the stringency of capital regulation, *Official supervisory power* is an index that measures whether supervisory entities have authority to take action to prevent and correct problems, *Private monitoring* is an index that measures whether there exists incentives/ability for the private monitoring of banks; *Regulation overall* is the first principal component of the four Barth, Caprio, and Levine (2013) indices, *Governance index* is the average of all six Kaufmann, Kraay, and Mastruzzi (2009) governance indicators; *Log GDP per capita* and annual real GDP growth come from the World Development Indicators; *Bank concentration* is the assets of the three largest banks as a share of all commercial banks' assets; *Bank credit* is the total private credit provided by the banking sector (log); *Real stock market return* is the 12-month average of the real stock market return; *Real exchange rate return* is the average annual real bilateral U.S. dollar exchange rate return; *Bilateral trade* is the maximum of bilateral imports or exports between two countries as a proportion of total imports by the target's country; *Growth opportunities* is the log of the inner product of the vector of global industry price-to-earnings ratios and the vector of country-specific industry weights, following Bekaert et al. (2007), and *Failed bank deals* is the sum of all failed bank M&A deals involving targets in country *i* in year *t* and all failed bank deals involving acquirers in country *j* in year *t*, as a proportion of all deals in target country *i* in year *t*. We take the average of each measure across the full sample period (1995–2012) for each country and report the average across all countries in the sample. We also report the average differences between acquirers and targets for each of these measures. All variables are defined in Appendix B. *Indicates significance at the 10% level.

Panel A: Descriptive Statistics of Country-Level Variables					
	<i>N</i>	Mean	Median	Maximum	Minimum
Restrictions on bank activities	78	7.18	7.19	12.00	3.22
Stringency of capital regulation	78	5.99	6.06	9.00	2.83
Official supervisory power	78	11.16	11.42	14.11	6.89
Private monitoring	77	8.21	8.00	11.00	5.72
Regulation overall	77	0.16	0.14	2.44	−1.32
Governance index	78	0.49	0.43	1.81	−1.18
Log GDP per capita	78	7.36	7.74	10.40	4.74
GDP growth	78	0.04	0.04	0.13	−0.01
Bank credit (log)	78	16.47	16.43	20.29	12.93
Bank concentration	78	0.65	0.67	1.00	0.00

(Continued)

Appendix C—Continued

Panel A: Descriptive Statistics of Country-Level Variables

	<i>N</i>	Mean	Median	<i>SD</i>	Maximum	Minimum
Real exchange rate return (%)	78	0.46	0.20	4.25	16.39	-21.80
Real stock market return (%)	78	9.18	9.58	10.70	47.71	-32.19
Bilateral trade	78	0.02	0.02	0.01	0.02	0.00
Growth opportunities	57	2.78	2.78	0.03	2.85	2.71
Cross-border ratio _{<i>i,j</i>} (%)	24,332	0.26	0	2.78	100	0
Δ restrictions on bank activities _{<i>j-i</i>}	23,732	0.01	0	2.96	9	-9
Δ stringency of capital regulation _{<i>j-i</i>}	22,842	0.05	0	2.63	8	-8
Δ official supervisory power _{<i>j-i</i>}	24,332	0.04	0	3.59	10.62	-10.62
Δ private monitoring _{<i>j-i</i>}	23,253	0.05	0	1.90	6	-5
Δ regulation overall _{<i>j-i</i>}	22,651	0.02	0.02	1.52	5.44	-5.44
Δ governance index _{<i>j-i</i>}	24,332	0.08	0.08	1.20	3.02	-3.02
Δ log GDP per capita _{<i>j-i</i>}	24,332	0.20	0.11	1.83	4.56	-4.56
Δ GDP growth (%) _{<i>j-i</i>}	24,332	-0.14	-0.17	4.27	20.93	-20.93
Δ bank concentration _{<i>j-i</i>}	24,332	0.77	1.04	31.44	99.74	-99.74
Δ bank credit _{<i>j-i</i>}	24,332	0.20	0.23	2.62	11.11	-11.11
Δ real exchange rate return (%) _{<i>j-i</i>}	24,332	-0.11	0.00	17.54	121.65	-277.62
Δ real stock market return (%) _{<i>j-i</i>}	24,332	-0.08	0.09	46.55	554.06	-458.88
Same language	24,332	0.11	0	0.31	1	0
Colony	24,332	0.03	0	0.18	1	0
Contiguous	24,332	0.04	0	0.20	1	0
Distance	24,332	8.58	8.98	0.95	9.89	5.15
Bilateral trade	24,332	0.03	0.01	0.06	0.90	0
Δ growth opportunities _{<i>j-i</i>}	24,332	0.00	0.00	0.06	0.36	-0.36
Failed bank deals	24,332	0.29	0.08	1.14	60	0

(Continued)

Appendix C—Continued

Panel B: Correlations Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
(1)	1																			
(2)	-0.007	1																		
(3)	0.005	0.043*	1																	
(4)	-0.002	0.199*	0.153*	1																
(5)	0.004	0.034*	0.017*	0.096*	1															
(6)	-0.008	0.458*	0.222*	0.728*	0.451*	1														
(7)	0.011	-0.495*	-0.053*	-0.288*	0.055*	-0.305*	1													
(8)	0.013*	-0.477*	-0.054*	-0.262*	0.063*	-0.281	0.915	1												
(9)	-0.014*	0.263*	0.020*	0.117*	0.052*	0.228*	-0.317	-0.377*	1											
(10)	-0.017*	-0.354*	-0.012	-0.227*	-0.033*	-0.232*	0.423	0.335*	-0.094*	1										
(11)	0.045*	-0.221*	0.044*	-0.176*	0.245*	-0.106*	0.398	0.452*	-0.197*	0.022*	1									
(12)	0.001	0.010	-0.034*	0.030*	0.019*	-0.041*	-0.016	0.024*	-0.045*	0.012	0.002	1								
(13)	0.000	0.004	0.014*	0.030*	-0.091*	-0.006	-0.035	-0.048*	0.137*	0.025*	-0.179*	0.020*	1							
(14)	0.078*	0.034*	0.015*	-0.004	-0.011	0.006	-0.032	-0.029*	0.018*	-0.007	-0.027*	-0.007	0.006	1						
(15)	0.073*	-0.008	0.009	-0.003	-0.006	-0.006	0.006	0.003	-0.001	0.003	0.001	0.001	0.002	0.299*	1					
(16)	0.116*	-0.008	-0.003	-0.003	-0.005	-0.006	-0.009	-0.013*	0.004	-0.003	-0.002	0.002	-0.003	0.192*	0.097*	1				
(17)	-0.106*	0.021*	0.001	0.023*	-0.023*	0.019*	-0.018	-0.021*	0.016*	-0.018*	-0.001	-0.023*	0.017*	-0.070*	-0.031*	-0.410*	1			
(18)	0.139*	-0.0067	0.014*	-0.021*	0.056*	-0.005	0.096	0.122*	-0.021*	-0.121*	0.280*	0.009	-0.030*	0.150*	0.088*	0.406*	-0.318*	1		
(19)	0.011	-0.110*	0.015*	0.002	-0.031*	0.000	0.272	0.199*	-0.085*	0.090*	0.069*	-0.092*	0.045*	0.001	-0.002	-0.005	-0.010	0.042*	1	
(20)	-0.001	-0.045*	-0.010	-0.019*	0.062*	-0.005	0.179	0.198*	-0.020*	-0.085*	0.272*	-0.004	-0.008	0.009	0.022*	-0.015*	0.050*	0.225*	0.061*	1
Variable																				
(1)	Cross-border ratio _{ij} (%)																			
(2)	Δ restrictions on bank activities _{j-i}																			
(3)	Δ stringency of capital regulation _{j-i}																			
(4)	Δ official supervisory power _{j-i}																			
(5)	Δ private monitoring _{j-i}																			
(6)	Δ regulation overall _{j-i}																			
(7)	Δ governance index _{j-i}																			
(8)	Δ log GDP per capita _{j-i}																			
(9)	Δ GDP growth _{j-i}																			
(10)	Δ bank concentration _{j-i}																			
	Δ bank credit _{j-i}																			
	Δ real exchange rate return _{j-i}																			
	Δ real stock market return _{j-i}																			
	Same language																			
	Colony																			
	Contiguous																			
	Distance																			
	Bilateral trade																			
	Δ growth opportunities _{j-i}																			
	Failed bank deals																			

REFERENCES

- Acharya, Viral V., 2003, Is the international convergence of capital adequacy regulation desirable? *Journal of Finance* 58, 2745–2781.
- Acharya, Viral V., Paul Wachtel, and Ingo Walter, 2009, International alignment of financial sector regulation, *Financial Markets, Institutions, and Instruments* 18, 188–190.
- Agarwal, Sumit, David Lucca, Amit Seru, and Francesco Trebbi, 2012, Inconsistent regulators: Evidence from banking, NBER Working paper 17736.
- Amel, Dean, Colleen Barnes, Fabio Panetta, and Carmelo Salleo, 2004, Consolidation and efficiency in the financial sector: A review of the international evidence, *Journal of Banking and Finance* 28, 2493–2519.
- Amihud, Yakov, Gayle L. DeLong, and Anthony Saunders, 2002, The effects of cross-border bank mergers on bank risk and value, *Journal of International Money and Finance* 21, 857–877.
- Anderson, James E., and Eric V. Wincoop, 2003, Gravity with gravitas: A solution to the border puzzle, *American Economic Review* 69, 106–116.
- Barth, James R., Gerard Caprio, and Ross Levine, 2006, *Rethinking Bank Supervision and Regulation: Till Angels Govern* (Cambridge University Press, Cambridge).
- Barth, James R., Gerard Caprio, and Ross Levine, 2013, Bank regulation and supervision in 180 countries from 1999–2011, NBER Working paper 18733.
- Beck, Thorsten, Asli Demirgüç-Kunt, and Ross Levine, 2006, Bank supervision and corruption in lending, *Journal of Monetary Economics* 53, 2131–2163.
- Beck, Thorsten, Asli Demirgüç-Kunt, and Ross Levine, 2009, Financial institutions and markets across countries and over time: Data and analysis, World Bank Policy Research Working paper 4943.
- Bekaert, Geert, Campbell R. Harvey, Christian Lundblad, and Stephan Siegel, 2007, Global growth opportunities and market integration, *Journal of Finance* 62, 1081–1137.
- Berger, Allen N., Robert De Young, and Gregory F. Udell, 2001, Efficiency barriers to the consolidation of the European financial services industry, *European Financial Management* 7, 117–130.
- Berger, Allen N., William C. Hunter, and Stephen G. Timme, 1993, The efficiency of financial institutions: A review and preview of research past, present, and future, *Journal of Banking and Finance* 17, 221–249.
- Bernard, Andrew B., Jonathan Eaton, Bradford J. Jensen, and Samuel Kortum, 2003, Plants and productivity in international trade, *American Economic Review* 93, 1268–1290.
- Blank, Sven, and Claudia M. Buch, 2007, The euro and cross-border banking: Evidence from bilateral data, *Comparative Economic Studies* 49, 389–410.
- Bouckaert, Jan, and Hans Degryse, 2006, Entry and strategic information display in credit markets, *Economic Journal* 116, 702–720.
- Bris, Arturo, and Christos Cabolis, 2008, The value of investor protection: Firm evidence from cross-border mergers, *Review of Financial Studies* 21, 605–648.
- Buccirossi, Paolo, Lorenzo Ciari, Tomaso Duso, Giancarlo Spagnolo, and Cristiana Vitale, 2011, Measuring the deterrence properties of competition policy: The competition policy indexes, *Journal of Competition Law and Economics* 7, 165–204.
- Buccirossi, Paolo, Lorenzo Ciari, Tomaso Duso, Giancarlo Spagnolo, and Cristiana Vitale, 2013, Competition policy and productivity growth: An empirical assessment, *Review of Economics and Statistics* 95, 1324–1336.
- Buch, Claudia M., and Gayle DeLong, 2004, Cross-border bank mergers: What lures the rare animal? *Journal of Banking and Finance* 28, 2077–2102.
- Buch, Claudia M., and Gayle DeLong, 2008, Do weak supervisory systems encourage bank risk-taking? *Journal of Financial Stability* 4, 23–39.
- Campa, José M., and Ignacio Hernando, 2006, M&A performance in the European financial industry, *Journal of Banking and Finance* 30, 3367–3392.
- Carbo-Valverde, Santiago, Edward J. Kane, and Francisco Rodriguez-Fernandez, 2012, Regulatory arbitrage in cross-border banking mergers within the EU, *Journal of Money, Credit, and Banking* 44, 1609–1629.

- Carletti, Elena, 2008, Competition and regulation in banking, in Anjan V. Thakor and Arnoud W.A. Boot, eds.: *Handbook of Financial Intermediation and Banking* (North Holland, London).
- Carletti, Elena, Philipp Hartmann, and Steven Ongena, 2013, The economic impact of merger control legislation, Working paper, Bocconi University.
- Carletti, Elena, and Xavier Vives, 2009, Regulation and competition policy in the banking sector, in Xavier Vives, ed.: *Competition Policy in Europe, Fifty Years of the Treaty of Rome* (Oxford University Press, Oxford, UK).
- Chor, David, 2009, Unpacking sources of comparative advantage: A quantitative approach, Working paper, Singapore Management University.
- Čihák, Martin, Asli Demirgüç-Kunt, Erik Feyen, and Ross Levine, 2012, Benchmarking financial systems around the world, World Bank Policy Research Working paper 6175.
- Claessens, Stijn, and Neeltje van Horen, 2014, Foreign banks: Trends and impact, *Journal of Money, Credit and Banking* 46, 295–326.
- Coffee, John C., 1999, The future as history: The prospects for global convergence in corporate governance and its implications, *Northwestern University Law Review* 93, 641–708.
- Cornett, Marcia M., and Hassan Tehranian, 1992, Changes in corporate performance associated with bank acquisitions, *Journal of Financial Economics* 31, 211–234.
- Correa, Ricardo, 2009, Cross-border bank acquisitions: Is there a performance effect? *Journal of Financial Services Research* 36, 169–197.
- Cybo-Ottone, Alberto, and Maurizio Murgia, 2000, Mergers and shareholder wealth in European banking, *Journal of Banking and Finance* 24, 831–859.
- Dell'Ariccia, Giovanni, and Robert Marquez, 2006, Competition among regulators and credit market integration, *Journal of Financial Economics* 79, 401–430.
- Djankov, Simeon, Caralee McLiesh, and Andrei Shleifer, 2007, Private credit in 129 countries, *Journal of Financial Economics* 84, 299–329.
- Doidge, Craig, G. Andrew Karolyi, and René M. Stulz, 2004, Why are foreign firms listed in the U.S. worth more? *Journal of Financial Economics* 71, 205–238.
- Dong, Hui, Frank M. Song, and Libin Tao, 2011, Regulatory arbitrage: Evidence from bank cross-border M&As, Working paper, University of Hong Kong.
- Duso, Tomaso, Klaus Gugler, and Burcin B. Yurtoglu, 2011, How effective is European merger control? *European Economic Review* 55, 980–1006.
- Duso, Tomaso, Damien J. Neven, and Lars-Hendrik Röller, 2007, The political economy of European merger control: Evidence using stock market data, *Journal of Law and Economics* 50, 455–489.
- Eaton, Jonathan, and Samuel Kortum, 2002, Technology, geography, and trade, *Econometrica* 70, 1741–1779.
- Ellis, Jesse, Sara B. Moeller, Frederik P. Schlingemann, and René M. Stulz, 2012, Globalization, governance, and the returns to cross-border acquisitions, NBER Working paper 16676.
- Erel, Isil, Rose C. Liao, and Michael S. Weisbach, 2012, Determinants of cross-border mergers and acquisitions, *Journal of Finance* 67, 1045–1082.
- Fally, Thibault, 2013, Structural gravity and fixed effects, Working paper, University of California, Berkeley.
- Focarelli, Dario, and Alberto F. Pozzolo, 2001, The patterns of cross-border bank mergers and shareholdings in OECD countries, *Journal of Banking and Finance* 25, 2305–2337.
- Giannetti, Caterina, Nicola Jentzsch, and Giancarlo Spagnolo, 2010, Information-sharing and cross-border entry in European banking, Centre for European Policy Studies Working paper 2990.
- Hagendorff, Jens, Michael Collins, and Kevin Keasey, 2008, Investor protection and the value effects of bank merger announcements in Europe and the U.S., *Journal of Banking and Finance* 32, 1333–1348.
- Hagendorff, Jens, Ignacio Hernando, Maria J. Nieto, and Larry D. Wall, 2012, What do premiums paid for bank M&As reflect? The case of the European Union, *Journal of Banking and Finance* 36, 749–759.
- Herrmann, Sabine, and Dubravko Mihaljek, 2013, The determinants of cross-border bank flows to emerging markets, *Economics of Transition* 21, 479–508.
- Hofstede, Geert, 1980, *Culture's Consequences—International Differences in Work-Related Values* (Sage, London).

- Houston, Joel F., Chen Lin, and Yue Ma, 2012, Regulatory arbitrage and international bank flows, *Journal of Finance* 67, 1845–1895.
- Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi, 2009, Governance matters VIII: Aggregate and individual governance indicators 1996–2008, World Bank Policy Research Working paper 4978.
- Köhler, Matthias, 2009, Transparency of regulation and cross-border bank mergers, *International Journal of Central Banking* 5, 39–74.
- Irrarrazabal, Alfonso, Andreas Moxnes, and Luca D. Oromolla, 2013, The margins of multinational production and the role of intra-firm trade, *Journal of Political Economy* 121, 74–126.
- La Porta, Rafael, Florencio Lopez-de-Silanes, and Andrei Shleifer, 2002, Government ownership of banks, *Journal of Finance* 57, 265–301.
- Laeven, Luc, and Ross Levine, 2009, Bank governance, regulation and risk taking, *Journal of Financial Economics* 93, 259–275.
- Laeven, Luc, and Fabián Valencia, 2012, Systemic banking crises database: An update, IMF Working paper 12/163.
- Levine, Ross, 2012, The governance of financial regulation: Reform lessons from the recent crisis, *International Review of Finance* 12, 39–56.
- Mayer, Thierry, and Soledad Zignago, 2011, Notes on CEPII's distances measures: The GeoDist database, CEPII Working paper 2011–25.
- Morrison, Alan D., and Lucy White, 2009, Level playing fields in international financial regulation, *Journal of Finance* 64, 1099–1142.
- Ongena, Steven, Alexander Popov, and Gregory F. Udell, 2013, When the cat's away the mice will play: Does regulation at home affect bank risk taking abroad? *Journal of Financial Economics* 108, 727–750.
- Pilloff, Steven J., and Anthony M. Santomero, 1998, The value effects of bank mergers and acquisitions, in Yakov Amihud, and Geoffrey Miller, eds.: *Bank Mergers and Acquisitions* (Kluwer Academic Publishers, Dordrecht).
- Ramondo, Natalia, 2008, Size, geography, and multinational production, Working paper, University of Texas at Austin.
- Rossi, Stefano, and Paolo F. Volpin, 2004, Cross-country determinants of mergers and acquisitions, *Journal of Financial Economics* 74, 277–304.
- Santos Silva, Joao M.C., and Silvana Tenreyro, 2006, The log of gravity, *Review of Economics and Statistics* 88, 641–658.
- Santos Silva, Joao M.C., and Silvana Tenreyro, 2011, Further simulation evidence on the performance of the Poisson pseudo-maximum likelihood estimator, *Economics Letters* 112, 220–222.
- Schwert, William G., 2000, Hostility in takeovers: In the eyes of the beholder? *Journal of Finance* 55, 2599–2640.
- Simonovska, Ina, and Michael E. Waugh, 2014, The elasticity of trade: Estimates and evidence, *Journal of International Economics* 92, 34–50.
- Starks, Laura T., and Kelsey D. Wei, 2013, Cross-border mergers and differences in corporate governance, *International Review of Finance* 13, 265–297.
- Stulz, René M., 1999, Globalization, corporate finance, and the cost of capital, *Journal of Applied Corporate Finance* 12, 8–25.
- Tenreyro, Silvana, 2007, On the trade impact of nominal exchange rate volatility, *Journal of Development Economics* 82, 485–508.
- Vander Vennet, Rudi, 2003, Cross-border mergers in European banking and bank efficiency, in Heinz Herrmann, and Robert Lipsey, eds.: *Foreign Direct Investment in the Real and Financial Sector of Industrial Countries* (Springer, Berlin).

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.